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Screen printing for silicon photovoltaics at terawatt scale: Trends, challenges, and opportunities

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ABSTRACT

Screen printing has been the dominant metallization technology for silicon solar cells since the 1980s. Within two decades, printed finger widths were reduced from $\sim$100 μm to $\sim$10–15 μm in mass production and $\sim$5 μm at R&D level. This review analyzes the economic drivers, technological catalysts, and milestones that have shaped screen-printed solar cell metallization. We first examine the proposed goal of < 2 mg/W_p silver consumption for sustainable terawatt-scale PV production by calculating various scenarios with updated assumptions. We derive a two-phase learning curve for finger width vs cumulative PV capacity, with learning rates of 12% (2008–2012) and 29% (2013–2024). We show how silver price volatility (peaks in 2008 and 2011) and interconnection transitions from three busbars to multi-busbar/multi-wire concepts catalyzed rapid fine-line innovations, reducing silver consumption while increasing cell performance. We discuss disruptive innovations such as knotless fine-mesh screens, laser-structured polyimide screens, and laser-enhanced contact optimization and their impact on fine-line printing. We quantify the learning rate of silver's module-price share, finding a 21.7% reduction per capacity doubling. However, silver's module-price share has risen since 2024 due to growing market share of TOPCon and rising silver prices. Finally, we outline pathways toward sustainable low-silver PV metallization, including stencil printing as an alternative to screen printing and the growing role of copper-, nickel-, or aluminum-containing pastes.

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I. INTRODUCTION

Over the last few decades, photovoltaics has evolved from a niche application to a mass-market technology. In 2024, the cumulative installed global photovoltaic (PV) capacity surpassed 2.2 TW_p.[1] For 2025, a further increase in ~650–700 GW of PV capacity is expected,[2] with latest forecasts predicting a moderate slowdown in PV production capacity growth for the first time in many years.[3] The impressive growth in global PV production over the last decade, especially in China, is leading to an increasingly serious challenge regarding the sustainable use of critical resources.[4–6] Silver, indium, and bismuth are currently considered the most critical resources for the PV industry.[7,8] In early 2026, particularly silver has experienced a dramatic price increase, peaking at over $110/oz[9] before temporarily retreating from those heights, thus highlighting the metal's pronounced price volatility. This dramatic price increase has been driven by a combination of geopolitical tensions, constrained mine supply, growing industrial demand—particularly form the automotive sector and the rapid built-out of artificial intelligence (AI) infrastructure[10]—as well as speculative activity in commodity markets. Since silver is currently a hard-to-replace raw material for metallization with an immense consumption of ~6150 t in 2024,[11] this strong price increase puts the PV industry under enormous pressure to act quickly and drastically to reduce silver consumption.[12] Over the last decade, and particularly since 2020, the annual share of PV in the total global silver supply has grown markedly from 10.3% (2020)[7] to 19.4% (2024).[11] As of autumn 2025, PV's share of total industrial silver demand was predicted to grow further to as much as 29%–41% by 2030.[13] Figure 1 shows the development of the average annual silver price and the annual silver demand of the PV industry for the period 2005–2025. The strong increase in both parameters, particularly since 2020, highlights the increasingly critical situation for the PV industry. An important factor behind the sharp increase in silver consumption since 2020 is the rapidly growing market share of high-efficiency cell concepts, such as tunnel oxide passivated contact (n-TOPCon) and silicon heterojunction (SHJ) solar cells. In particular, TOPCon—with a market share of almost 70% in 2025—has largely superseded the passivated emitter and rear contact (PERC) solar cell as the dominant cell architecture.[14] Due to the bifacial cell structure and the need for silver contacts on both sides, TOPCon solar cells have a silver consumption that is around 40%–60% higher compared to PERC solar cells, depending on cell design and measurement basis.[14,15] It is even suspected that silver might have overtaken polysilicon as the single largest cost component and could currently account for about 30% of total cell costs and 16%–17% of module costs.[16]

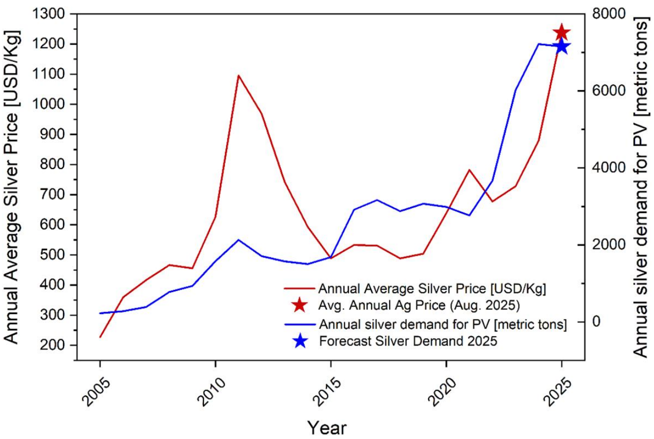

FIG. 1. Development of the annual average silver price and the annual total demand for the PV industry from 2005 to 2024, including prediction for 2025 based on Ref. 8. Annual silver price averages computed from monthly World Bank Commodity Price Data ("Pink Sheet") silver prices (USD per troy ounce);[17] converted to USD per kilogram using 1 kg = 31.103 476 8 troy ounces (own calculation).

This raises the question of what level of silver consumption is sustainable in the long term for PV production at the terawatt scale. To answer this question, a comprehensive study has been carried out by Zhang *et al.* in 2021[7] considering the results of earlier studies.[18,19] The authors proposed a maximal tolerable silver consumption of less than 2 mg silver per W_p for a long-term sustainable PV production with an assumed production volume of 3 TW/a.[7] However, the drastically increasing market share of silver-intensive, high-efficiency cell concepts like TOPCon and SHJ in the last few years has exacerbated the situation, suggesting that this target may no longer be current and may need to be revised downward. This impression is reinforced by a comprehensive study conducted by Hallam *et al.* in 2023,[5] which shows that at the current ~20% silver learning rate, PV deployment compatible with net-zero by 2050 would require most or even all known silver reserves (especially if silver-intensive cell concepts dominate). The study implies that efforts to raise the silver learning rate to around 30% via rapid deployment of silver-lean and silver-free metallization must be drastically accelerated. To assess the validity of the proposed limit of 2 mg/W_p under current conditions, we perform a two-dimensional parameter study of the key influencing parameters and calculate different scenarios within Sec. III A.

Periods of sharp increases in the price of silver—such as in 2011 and again in 2025—have continuously pressured the PV industry to rapidly reduce silver consumption through fast innovation cycles in metallization and interconnection. Rapid optimization of materials and machinery and disruptive developments in the field of screen printing and interconnection technology enabled a drastic reduction of silver consumption while mitigating electrical and optical losses. For 2024, the average silver consumption can be estimated with ~13 mg/W_p, based on actual figures for annual global fabricated PV capacity[20,21] and the PV industry's silver consumption in 2024,[11] as shown in Sec. III C. Various R&D results obtained, for example, by the University of New South Wales (UNSW),[15] ODTÜ-Günam,[22] and Fraunhofer ISE[23] in 2025 demonstrate that high-efficiency solar cells with silver consumption below 2 mg/W_p are already achievable by replacing silver with alternative materials. Yet, transferring these encouraging R&D results into industrial mass production still seems very challenging in terms of reproducibility, electrical performance, and long-term stability on module level.

According to the actual ITRPV report,[14] flatbed screen printing remains the near exclusive technology for industrial-scale metallization of silicon solar cells. Various alternative metallization approaches, such as electroplating,[24–27] stencil printing,[28] parallel dispensing,[29] FlexTrail,[30] rotary screen printing,[31] laser-induced forward transfer (LIFT),[32] simultaneous foil transfer metallization using a newly developed approach by Lumet Ltd.,[33] pattern transfer printing (PTP),[34] and string printing[35] have emerged over the last few decades. While some of these new approaches have shown promising results for fine-line printing and silver reduction, screen printing has remained the dominant metallization method due to its ease of use, reliability, and cost-effectiveness. New generations of screens and pastes have enabled the printing of ever-finer front-side contacts on the solar cell, which contributed significantly to higher efficiencies and lower silver consumption.[28,36,37]

Within this work, we provide four key contributions: First, we review fundamentals of solar cell metallization and important technical milestones over the past decades. Based on published results for screen and stencil printing from 2008 to 2025, we compile an updated dataset and derive a two-phase learning curve for printed finger width vs cumulative global PV capacity, revealing a notable acceleration of the learning rate in the last decade. Second, we reassess the proposed limit of 2 mg/Wp silver consumption using a two-dimensional parameter study of the key input parameters. We further quantify the learning rate of silver's share of module price since 2010 to assess the impact of costs related to silver. Third, we show how interconnection and process-related innovations—including laser-enhanced contact optimization (LECO) and the introduction of polyimide (PI), knotless, and back-etched meshes—have accelerated fine-line printing. Finally, we outline practical routes to achieve minimal silver consumption at the module level, including fine-line screen and stencil printing and the use of alternative materials such as copper and nickel pastes.

II. FUNDAMENTAL ASPECTS AND REQUIREMENTS FOR SOLAR CELL METALLIZATION AND INTERCONNECTION

A. Fundamental requirements of solar cell electrodes

In most solar cell concepts, metallic electrodes are applied to the front and back of the solar cell to extract the photogenerated charge carriers from the semiconductor. An exception are back-contact solar cells such as the so-called interdigitated back contact (IBC) solar cell,[38,39] where the metallic contacts are applied exclusively on the rear surface. In essence, solar cell metallization must meet various requirements regarding electrical and optical performance at the cell level, as well as performance and reliability at the module level. The most important aspects include the following:

- Minimization of shading losses on the active cell surfaces on both the front and rear sides in the case of bifacial cell concepts[40]
- Minimization of the series resistance contribution due to the resistance of the front and rear contact grids. In addition to the specific conductivity of the contacts, this is strongly influenced by the metallization layout (number of contact fingers) and the interconnection concept of the cells during module fabrication[41,42]
- Sufficiently low contact resistivity ρ_c of metal–semiconductor contacts[40]
- Low recombination velocity (metal-related saturation current density $j_{0,met}$) at the metal contacts[43]
- Sufficient adhesion of the connecting ribbons or wires by soldering, electric conductive adhesives (ECA), or other means of interconnection[44]

The optimal design and electrical performance of the metallic electrodes, as well as the selected interconnection concept, have a strong influence on the resulting electrical performance and efficiency of the cell and module. It is therefore essential that these aspects, as well as the technical limits of the chosen metallization process, are considered and optimized to achieve an optimal result at the cell and module level. Another important aspect for industrial production is profitability and productivity, which are strongly influenced by factors such as material costs (e.g., screens and metallization pastes), throughput, and the lifetime of the printing form (screen or stencil). Therefore, it is essential to find a compromise between cell performance and cost-effectiveness. The specific design of the metallization step therefore always represents a complex trade-off between technical and economic aspects.

B. Optimizing the optical and electrical properties of the electrodes

The following paragraph is based on the general description of requirements for solar cell metallization by Tepner and Lorenz.[42] The front-side metallization should ensure an optimal trade-off between shading losses of the metal grid (finger shape and width, number of contact fingers), contribution of the grid to the overall series resistance (grid resistance and contact resistance), and the total silver laydown per cell (small finger width and height, uniformity of fingers). To achieve this, the grid layout must be individually adapted depending on the selected interconnection concept, the material parameters of the semiconductor, and the technical limits of the selected metallization technology. To minimize optical shading and reflection losses in the encapsulated module, the metallization grid should enable the solar cells to capture as much of the incoming light as possible. Considering the individual contact, the three-dimensional shape of the contact fingers (taking into account the actual reflection losses in the module) plays an important role (Fig. 2). The actual effective or shading-relevant finger width $w_{f,eff}$ in the encapsulated state can deviate significantly from the geometric finger width w_f depending on the shape of the fingers as shown by Blakers,[45] Stuckings and Blakers[46] and Woehl.[47] In an ideal case, the contact fingers reflect a substantial part of the incoming light directly onto the active cell area or indirectly via total internal reflection on the module glass surface.[45,47] In theory, a triangular finger geometry would ideally reflect the incident light completely back onto the active cell surface and thus realize an "effectively transparent" contact grid.[48] The feasibility of such a triangular finger geometry was demonstrated with several metallization approaches including stencil printing (Fig. 2).[35,48–51]

From an electrical perspective, the series resistance contribution $r_{s,grid}$ plays an important role in achieving a high fill factor (*FF*) of the cell, as described in Refs. 42, 53, and 54. The optimal compromise between shading losses and series resistance contribution is strongly influenced by the number of busbars/wires and contact fingers within the grid layout[55] and can be adjusted using suitable simulation tools.[56] The maximum acceptable finger resistance per unit length R_L for a defined series resistance contribution limit $r_{s,f}$ of the fingers can be calculated for a given interconnection scheme and grid layout according to Refs. 53, 54, and 57. Typical thresholds for common interconnection schemes are provided in Ref. 42. This clearly indicates the strong dependence of the acceptable grid (and finger) resistance on the selected interconnection scheme. Decreasing the distance between

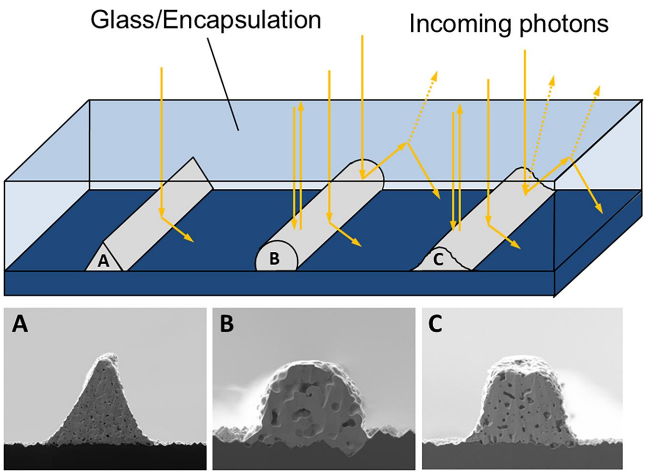

FIG. 2. Schematic showing reflection behavior of finger shapes under ethylene vinyl acetate (EVA)/glass encapsulation, with scanning electron microscopy (SEM) images realized at Fraunhofer ISE: (a) triangular finger profile via glass-foil stencil printing;[51] (b) circular finger profile via multi-nozzle parallel dispensing;[29,52] (c) parabolic/Gaussian profile via screen printing. Adapted with permission from Tepner and Lorenz, Prog. Photovoltaics **31**(6), 557–590 (2023). Copyright 2023 Authors, licensed under a Creative Commons Attribution 4.0 International License (CC BY 4.0).[42]

wires allows conduction-relevant cross-sectional area of the fingers, and hence the finger width, to be reduced. This, in turn, permits closer finger spacing at a given optical shading level and thus reduces lateral resistance in the solar cell. Developing new interconnection concepts like multi-wire approaches was therefore an essential prerequisite to boost new developments in screen printing toward printing ever narrower contacts with less silver consumption.

C. A brief history of solar cell metallization

The following paragraph is a modified and updated version of the historical review previously published by Tepner and Lorenz.[42]

The idea to use screens for printing of conductive circuits on electronic components dates back to the first half of the 20th century and, according to various sources, is commonly credited to Paul Eisler, widely regarded as the inventor of the so-called printed circuit board (PCB).[58–60] However, many decades were to pass before screen printing was used for solar cell metallization. Until the mid-1970s, the production of silicon solar cells was a highly specialized, non-automated sequence of processes, used primarily for aerospace applications and characterized by very low annual production volumes. The metallization of these early solar cells was mainly achieved by more complex and costly processes such as electroplating, vapor deposition, or sputtering of subsequent metal layers (e.g., Ti/Pd/Ag) using a photolithographic mask.[61–63]

In the mid-1970s, the growing use of PV for terrestrial applications and the resulting increase in production volume created a strong need for cost reduction and thus accelerated the development of more efficient silicon solar cell production processes.[64] As a result, established production methods from other branches of electronics manufacturing were gradually transferred to the manufacturing process of silicon solar cells. Spectrolab's development of the first screen-printed aluminum back-surface-field solar cell (Al BSF) in the mid-1970s can be seen as the starting point of industrial screen printing for solar cell metallization.[65] Throughout the 1980s, Al BSF quickly became the predominant industrial concept and dominated global PV production for more than 30 years.[66] Typical H-pattern cells at this time were metalized with broad contacts around 120–150 μm in width.[67] Due to the low annual PV production volume at that time, silver consumption concerns did not play a role at all. Another important milestone was the development of silver-based fire through pastes which provided low contact and lateral grid resistance, avoided penetrating the shallow p–n junction, and ensured good adhesion after firing.[68] By the early 1980s, screen printing had already established itself for the application of the metal electrodes for solar cells in industrial production.[69]

Research activities in the 1990s focused on further optimizing the screen printing process to achieve finer contacts in combination with selective emitters.[70] Furthermore, the paste rheology[71] and the influence of the surface texture[72] were increasingly investigated. In the early 2000s, the physical and chemical contact-forming mechanisms of firing-through silver pastes on n-type emitters, as well as the microstructure of the metal–semiconductor contact interface, received significant attention in the research community.[73,74] Schubert *et al.* proposed a comprehensive model to explain the chemical and physical contact formation as well as the current transport mechanisms between the screen-printed silver contacts and the n-type emitter.[75–77] Further studies deeply investigated the contact formation mechanism within the firing process,[78–89] the influence of the glass-frit composition,[90–95] as well as the special characteristics of the metal–semiconductor contact at the microstructure level.[82,92,94,96,97]

The Al BSF solar cell was the undisputed and predominant industrially manufactured cell concept until about 2012–2013. At this time, the technical possibilities for further optimization of this cell concept had been exhausted and the cell efficiency stagnated at a level of ~20% on Czochralski-grown (Cz) silicon.[98] The development of the PERC solar cell concept in the late 1980s based on the work of Blakers *et al.*[99] and previous conceptual considerations[100] was another important milestone in the history of industrial PV mass production. By introducing an additional dielectric passivation layer on the back between silicon and aluminum (Al) metallization, recombination losses were effectively reduced. In parallel, the increased internal reflection of long-wavelength light at the rear dielectric interface was improved by the full-area Al rear-side metalization,[98,101] which was originally realized by evaporation[102] until screen-printed Al metallization with laser-fired local contacts (LFC) was developed.[103] A further breakthrough was the development of a fire-stable silicon nitride (SiN_x) passivation and diffusion barrier layer, which was opened locally and metalized by screen printing.[104] Finally, a process sequence for laser contact opening of the rear passivation (LCO) followed by Al screen printing metallization and a high-temperature co-firing process became the standard process for industrial PERC solar cells.[101] A comprehensive overview of the functional principle, process routes, and specific aspects of PERC and related PERx variants is provided by Preu *et al.*,[101] Dullweber *et al.*,[105] and Green.[106]

The industrial mass production of PERC solar cells started in 2012 with SolarWorld being the first to produce PERC cells on industrial level.[107] In the following years, the PERC cell concept was quickly adopted by Chinese PV manufacturers and rapidly replaced the Al

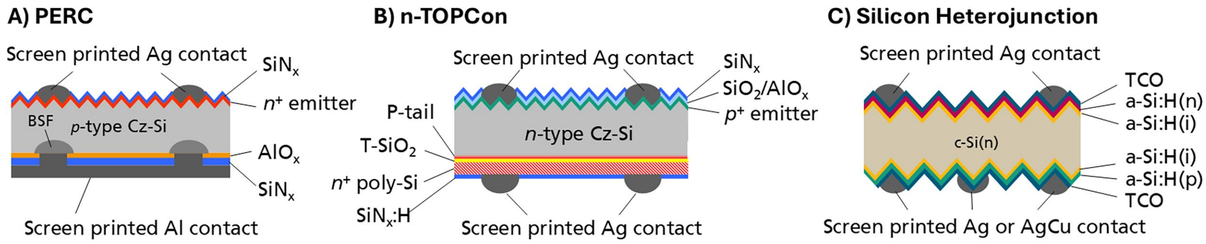

FIG. 3. Schematic of relevant solar cell concepts: (a) p-type passivated emitter and rear contact solar cell (PERC), (b) n-type tunnel oxide passivated contact solar cell (n-TOPCon), (c) silicon heterojunction solar cell (SHJ).

BSF solar cell as the dominant solar cell technology. With an impressive average annual efficiency improvement of approximately 0.6% points per year,[98] the PERC cell concept rapidly became the predominant cell concept and manufacturers were finally achievable efficiencies of more than 23% through constant further development.[108]

In recent years, high-efficiency solar cell concepts with carrier-selective contacts (also called "passivating contacts") have increasingly entered industrial production and have gradually begun to replace the prevailing PERC cell concept [Fig. 3(a)].[109] A comprehensive overview regarding carrier-selective contacts is provided by Refs. 98 and 110. These cell concepts effectively suppress surface recombination at the metal contacts (dark saturation current density $j_{0,met}$)[111] and thus enable silicon-based solar cells to approach the theoretical efficiency limit of 29.4%[112] to be approached for silicon-based solar cells. To date, two high-efficiency cell concepts have broadly established themselves in industrial PV mass production and gradually replaced the PERC cell concept:

- In 2013, Feldmann *et al.* published the first solar cell featuring a boron-diffused emitter and a passivating rear contact.[113,114] The tunnel oxide passivated contact solar cell concept [n-TOPCon, Fig. 3(b)], originally developed at Fraunhofer ISE building on earlier work[115–120] focused on effectively reducing recombination at the metal–semiconductor interface by introducing an intermediate passivation layer. Further related cell concepts based on the same concept like poly-Si on oxide (POLO),[121] monopoly,[122] and PERPoly[123] were developed in the following. TOPCon proved to be particularly suitable for industrial mass production due to the compatibility with established high-temperature processes, a comparatively easy integration into existing PERC production lines and an excellent surface passivation quality enabling a very high conversion efficiency level up to more than 27%.[124,125]
- The *silicon heterojunction* (SHJ) solar cell concept[126] using intrinsic and p-/n-type hydrogenated amorphous silicon (a-Si:H) layers for surface passivation [Fig. 3(c)], originally developed in 1983[127] based on previous work of Fuhs *et al.* on a-Si:H/c-Si heterostructures[128] and first commercialized by Sanyo.[129] Due to the temperature sensitivity of the a Si:H layers, the maximum processing temperature for SHJ solar cells is limited to approximately 200–230 °C,[130] which requires a low-temperature curing process using specific paste compositions for the screen-printed metallization. A world-record efficiency of 27.81% at that time was demonstrated as of late 2025 by combining the SHJ cell concept with a back-contact cell architecture.[131]

Various developments have contributed to a fast efficiency increase in industrially produced n-TOPCon solar cells in recent years. A particularly important technological milestone was the development of the so-called laser-enhanced contact optimization (LECO) process (Fig. 4) by CE Cell Engineering GmbH in 2016,[132] which quickly became widespread in industry in the following years.

By scanning a laser across the cell while applying a reverse-bias voltage, LECO treatment facilitates nondestructive carrier injection, thereby substantially improving the silver-silicon contact while preserving contact passivation.[135,136] By improving the contact resistance of cells fired with a lower thermal budget, LECO enables a decrease in metal-induced recombination mainly on the boron emitter side and to minor degree on the n-TOPCon side.[133] Krassowski *et al.* demonstrated that LECO treatment enables a substantial increase in efficiency due to additional gains in short-circuit current and open-circuit voltage on cell level.[136] In addition, LECO enabled the substitution of previously used Al-containing silver pastes (AgAl) with pure silver pastes for metallization of the p^+-emitter on the front side.[136] This, in turn, had a major impact on the fine-line printability of these paste for front-side metallization, as the elimination of large Al particles made it possible to develop silver pastes specifically optimized for printing much finer contacts. Silver pastes also enabled a lower contact resistance to the p^+-boron emitter, as well as a lower bulk resistance and thus a more effective use of silver.[137,138] The absence of

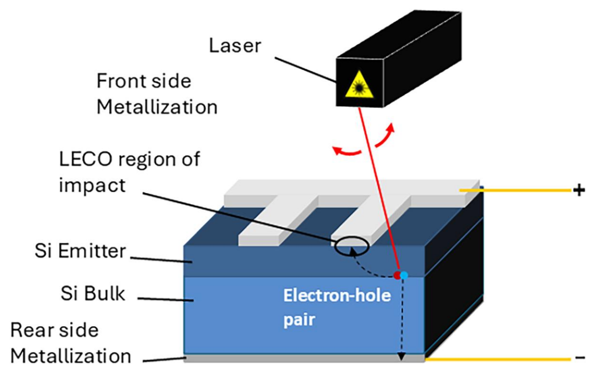

FIG. 4. Schematic of LECO process: A laser scans the front side of the solar cell under reverse bias, injecting carriers and driving a high local current through the metal–semiconductor interface. This locally reduces the contact resistivity across the wafer and thereby boosts the fill factor.[133] Adapted with permission from Krassowski *et al.*, Sol. RRL **6**(5), 2100537 (2022). Copyright 2022 Authors, licensed under a Creative Commons Attribution 4.0 International License (CC BY 4.0).[134]

Al particles (and associated spiking into the p^+-emitter) further reduces contact recombination ($J_{0,met}$), which is reflected in a higher V_{oc} and *pFF*, especially when combined with LECO processing.[133] The application of the LECO process in combination with advanced fine-line screen printing also contributed substantially to latest efficiency world-record achievements like the record efficiency of $\eta_{max} = 27.79\%$ on a bifacial n-TOPCon solar cell demonstrated by Jinko Solar in 2025.[124] At the same time, great progress has been made regarding the formulation of front-side silver paste for contacting high-Ohmic p^+-emitters with very low metal-induced recombination.[139] Using latest fine-mesh screen configurations, the front-side metallization for n-TOPCon solar cells is expected to reach $\sim$10 μm in industrial mass production.[37,139] On R&D level, contact fingers down to an even smaller finger width of only $\sim$5 μm using stencil printing have already been demonstrated.[37]

D. Next generation of high-efficiency solar cells

Currently, the most intensively discussed next generation of solar cells beyond silicon single-junction devices comprises architectures that incorporate an active perovskite layer. These include perovskite single-junction cells as well as silicon–perovskite[140] and perovskite–perovskite tandem solar cells. All of these device concepts feature a perovskite (sub-)cell on the illuminated front side, which imposes specific requirements on the metallization process.[141–143]

While the metallization of tandem solar cells is currently mainly realized by evaporation, the scale-up of tandem solar cells for industrial mass production requires efficient and productive methods for the metallization. The most promising approach for industrial mass production is again screen printing, due to its well demonstrated reliability, high productivity, and cost-efficiency.[42,143] For screen-printed contacts, low-temperature pastes based on Ag or AgCu particles are considered the most promising option,[142,143] as substantial process and materials know-how from heterojunction silicon technology can be leveraged. In order to prevent degradation of the perovskite absorber and adjacent transport layers, several constraints must be met: (i) the curing temperature must be reduced to sufficiently low values (state-of-the-art pastes can be cured at approximately 100–150 °C within a few minutes),[141,143] (ii) chemically non-aggressive solvents must be employed,[144,145] (iii) metal ion migration into the perovskite absorber must be suppressed (typically, the TCO layer also functions as a barrier against metal diffusion into and halogen diffusion out of the perovskite),[146] and (iv) the metallization process must be mechanically gentle, with minimized squeegee pressure and shear forces, to avoid crack formation or even delamination of the perovskite layers.[147] In addition, the paste must exhibit sufficient adhesion to the perovskite cell to enable reliable interconnection and long-term stability, which remains a central challenge for perovskite devices due to the thermal budget required for standardized reliability testing.[142] In addition to conventional thermal curing, alternative low-thermal-budget sintering approaches such as intense pulsed light (IPL)[148,149] or laser sintering are promising options to further reduce thermal stress on the device stack.

To achieve high power conversion efficiencies in perovskite single-junction solar cells with screen-printed metallization, the requirements are in many respects comparable to those of silicon single-junction devices. However, achieving low bulk resistivity under strict low-temperature constraints is more challenging.[142,143] Therefore, metal nanoparticle-based pastes represent a particularly promising approach as they might provide a low bulk resistivity at substantially lower temperature.[143] In tandem solar cell applications, the requirements for bulk and contact resistivity are less stringent, as the operating current density is approximately half that of a single-junction cell. This relaxation enables the implementation of fine-line screen printing with large finger spacing, thereby significantly reducing optical shading losses while maintaining acceptable series resistance.[142,143]

E. Technological advances and milestones in screen printing and cell interconnection

In this section, we highlight the major technological innovations and milestones in screen printing and related cell interconnection technologies that have contributed to the drastic reduction of screen printable finger width, reduced silver consumption, and increased cell efficiency over the past 20 years. Figure 5 highlights key milestones and impact factors with respect to the evolution of solar cell metallization and interconnection.

A fundamental aspect closely linked to screen-printed metallization of solar cells is the chosen interconnection concept. The choice of the interconnection concept has a very strong influence on the design and the required electrical properties of the printed metallization grid. In the first decades of industrial PV production, the interconnection of the cells in strings was usually realized via two ribbons which were soldered onto the printed and fired busbars. This concept is commonly known as the "H-pattern" grid design. Around 2007,[150] the PV industry transitioned from two to three busbars/ribbons, which was motivated by several advantages:[151] (i) increased tolerance to finger interruptions by shortening finger segments; (ii) reduced ribbon current by a factor of $\sim$1.5, enabling narrower ribbons and fewer breakages; and (iii) a slight efficiency gain with a matched finger count.[151,152] Furthermore, this transition also enabled the usage of larger wafer formats which supported the transition from the former standard format 125 × 125 mm^2 to larger 6-in. wafers (156 × 156 mm^2) from around 2010 onward.[153] Due to the comparatively long finger segment lengths between two busbars, H-pattern grid designs required contact fingers with high lateral conductivity and thus a correspondingly large finger cross-sectional area. Contact fingers were printed with a typical finger width of approximately 80–120 μm at this time.[152,154–156] From 2010 onward, the price of silver rose rapidly in the wake of the financial crisis, peaking at around USD 1600/kg in April 2011.[157] This development induced a strong pressure to quickly reduce silver consumption, which accelerated the development of novel screens and pastes for printing finer contact fingers and at the same time intensified R&D activities to develop alternative processes such as the substitution of silver by electroplating of nickel and copper.[158,159] Through the continuous improvement of the screen printing pastes and the use of finer screens, finger width could be reduced in the following years, whereby the increasing finger resistance per unit length R_L (Ω/cm) was compensated by an adjustment of the grid layout (additional current paths by increasing the number of contact fingers). The increasing use of dual printing[160,161] enabled sequential printing of fingers and busbars using separately optimized pastes and screens. Thus, fingers could be printed with contacting (glass-frit containing) fine-line silver paste while busbars are printed in a separate step using a non-contacting silver paste ("floating

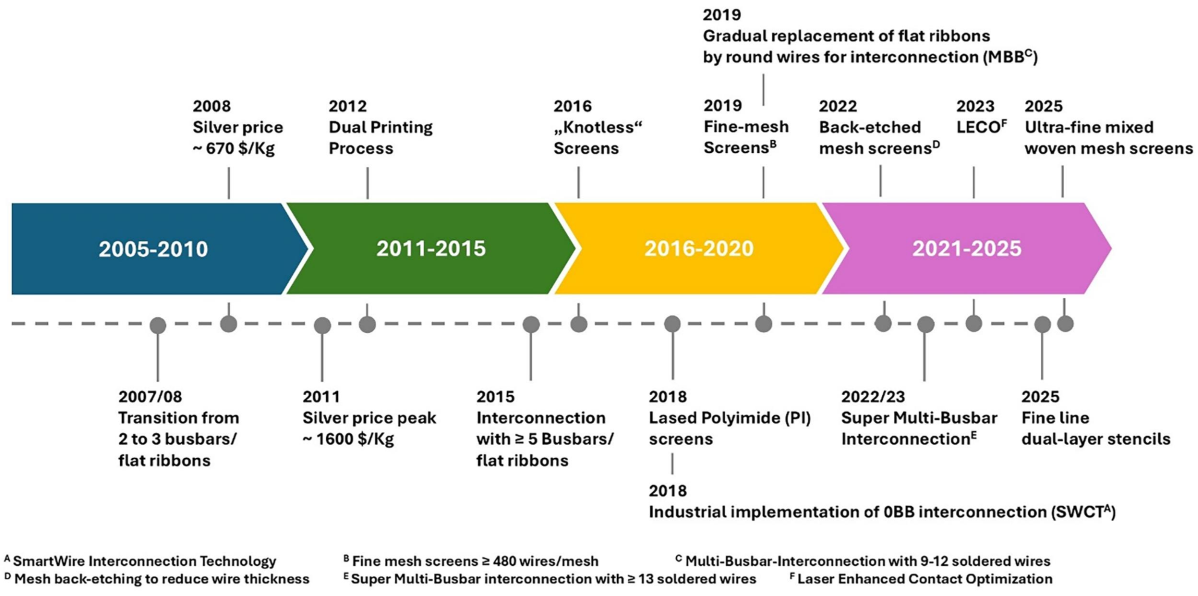

FIG. 5. Timeline of important milestones with respect to screen-printed solar cell metallization from 2005 to today. Since the development and industrial implementation of the mentioned technological advances often extent over a certain period of time, the mentioned dates should be understood as indications for the industrial establishment of the respective technology in the PV industry.

busbars," i.e., not fired through the passivation) to reduce the overall metal-related surface recombination.[42] Another advantage of this dual printing approach is the reduction in silver consumption, as the busbars can be printed with a different screen configuration optimized for a lower layer thickness, as well as special silver busbar pastes with reduced silver content.

The temporary trend of using a double-printing process (print on print)[162] to increase the aspect ratio and thus the lateral conductivity of narrow contacts is rarely used nowadays. New generations of pastes enable a sufficiently high aspect ratio, and therefore adequate lateral conductivity, even with a single printing step.

The trend toward fine-line printing was significantly accelerated when the industry began increasing the number of busbars or ribbons from three to four,[163] five,[164] and finally six[165] starting around 2015, thereby gradually reducing the finger segment lengths between two busbars (Fig. 6). By 2017, the so-called multi-busbar (MBB) technology using 9–15 soldered (round) wires instead of ribbons gained significant traction in the PV industry.[37,166] This approach enables an additional efficiency gain on the module level as the round-shaped wires reflect additional light onto the solar cell which would be lost in the case of a flat ribbon.[167] Additionally, contact fingers with reduced height can be printed as the considerably shorter current paths (finger segments) in between the wires prevent series resistance losses due to a higher finger resistance R_L. Since 2022, the so-called "super multi-busbar" concept (SMBB) with more than 16 up to 24 wires soldered on mini-busbars has been increasingly used for large cell format cells,[37] see Fig. 6.

At the same time, there is a trend toward reducing the cell area by cutting the large initial cell formats to smaller cell sizes. In recent years, cutting full cells into half or third cells has been widely

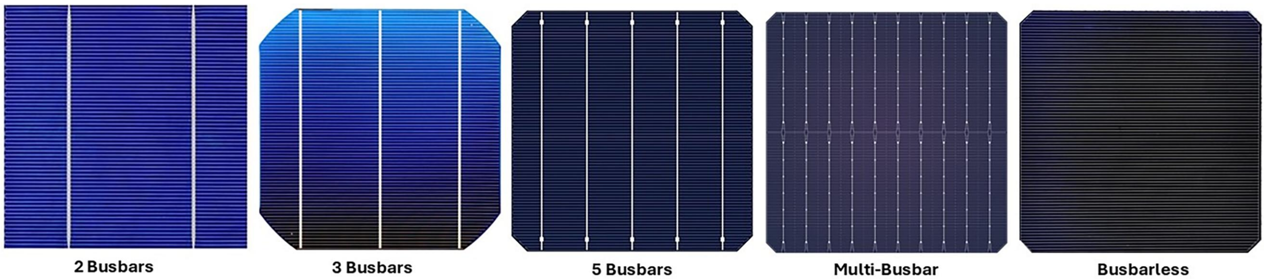

FIG. 6. Initially, solar cells were interconnected by two to three soldered ribbons soldered to printed busbars for current collection. Starting around 2015, the industry witnessed a rapid transformation regarding interconnection, expanding to 4, 5, and subsequently 6 busbars with narrow, soldered ribbons. In the last decade, multi-busbar (MBB) and super multi-busbar (SMBB) technologies have gained widespread adoption, employing soldered wires on printed mini-busbars rather than conventional ribbons, thereby delivering superior optical and electrical performance characteristics. In the case of busbarless interconnection concepts, the wires are directly soldered to the contact grid without printed busbars or pads using foil lamination with embedded wires coated with low-temperature solder alloys (e.g., SmartWire interconnection technology, integrated film connection) or adhesive gluing with pre-soldering interconnection.

established. Currently, large manufacturers such as Jinko are already working on the mass production of quarter cells, which offers several advantages: reduced resistive losses due to lower current per wire, reduced optical losses, a lower risk of hot spots, and a potential increase in module packing density (e.g., by shingling).[139] Today, soldered ribbon interconnection has been almost completely replaced by multi-wire interconnection methods with an estimated market share of ∼95% for MBB and SMBB interconnection.[14,139] Furthermore, 0BB (busbarless) interconnection concepts—e.g., integrated film connection (IFC), strip film interconnection, and adhesive gluing with pre-soldered interconnection[37]—are gaining importance, with a market share of approximately 5%–10% in 2025.[14,37,139] A detailed overview regarding various approaches for 0BB interconnection is provided in Ref. 139.

This rapid technological transition regarding the applied cell interconnection concepts had a strong impact on the development of screen printing for solar cell metallization. The new interconnection techniques made it possible to print significantly finer contacts due to a higher tolerable finger resistance R_L of the grid. Since this also had the potential to substantially reduce silver consumption, this evolution resulted in a great industrial need for the development of finer screens and suitable pastes for printing very fine contacts. Consequently, new generations of fine-mesh screens using higher mesh counts and finer wires evolved rapidly, while the development and iterative optimization of new fine-line pastes was strongly accelerated in parallel.

The most important technical innovations and milestones with respect to fine-line screen printing throughout the last decade are summarized in the following:

- A far-reaching innovation was the development of screens with a laser-patternable masking layer based on polyimide (PI),[168,169] which spread quickly after the market introduction in 2018 (Fig. 7). This technology rapidly and largely replaced the traditional screen fabrication based on UV-sensitive photopolymer emulsions or capillary films from 2019 onwards.[170] Polyimide as masking layer material has several clear advantages over traditional photo emulsions: It is much more resistant to mechanical abrasion, compatible with numerous solvents and can be structured very precisely using laser technology.[171] This innovative development thus represented a decisive step forward in both economic and technological terms. On the one hand, substantially finer (narrower) finger openings could be realized on screens using a laser process. On the other hand, the chemical resistance of PI against a wide range of solvents allows greater freedom in optimizing fine-line pastes. The most decisive advantage, however, is the substantially increased lifetime of the screens due to the high durability of PI. Such screens achieve a service life of several hundred thousand to 1×10^6 prints in industrial production today,[28,172] which substantially increases their lifetime compared with traditionally fabricated screens and thus significantly reduces costs.
- The ability of mesh manufacturers to develop advanced weaving processes for ultra-fine-mesh screens using very thin stainless steel or tungsten wires[36] with a wire diameter of down to 9 μm.[37] The development of chemical back-etching processes for the woven mesh enabled a substantial and cost-effective reduction of the initial wire diameter, thereby increasing the open area. Nowadays, it is possible to obtain a very small wire thickness down to 4 μm using this approach.[37,173] Newly developed back-etching processes of the wires by using the PI layer as a mask enable a localized reduction of the wire thickness which maintains the stability of the overall mesh. Since the effort of the weaving process and thus the mesh costs increase exponentially with decreasing wire thickness, the back-etching process has a major influence on the screen costs, especially for very fine meshes. Today, a wide range of fine-mesh screens with a mesh count of up to 760 wires per inch and a very thin wire thickness down to 4 μm are used in PV mass production,[37,173] see Fig. 8.
- A further groundbreaking innovation based on earlier concepts and patents was the introduction of so-called "knotless screens" or "zero-degree-angled screens" with a mesh angle of 0° instead of the commonly used 22.5° or 30° screens.[174] Using this approach, the contact finger openings are realized between the wires thus avoiding mesh intersections (knots) in the channels (Fig. 9). As a result, the open area in the finger channels is substantially increased, while mesh marks due to wire intersections can be effectively reduced. Using knotless screens, particularly fine contact fingers can be printed with substantially better uniformity and thus electrical performance at lower silver consumption compared to screens with traditional 22.5° or 30° mesh angle.[175,176] In the early years of knotless screen fabrication, the screen mesh had to be prepared by manually removing wires in the region of the designated channel openings. This was later improved by selective, laser-assisted removal of wires. Today, the use of image recognition and precision laser systems allows for precise placement of the laser-opened channels between the wires without the need for wire removal. This increases the stability of the screen considerably so that actual knotless screens can obtain a very high lifetime of up to 800 000–1 000 000 prints.[37]
- Finally, a deep understanding of the relevant paste properties and the impact of the paste-screen interaction formed the basis for rapid development of high-performance fine-line metallization pastes. Xu *et al.*[177] as well as Tepner *et al.*[178] identified the wall slip behavior of the paste in the screen finger channel

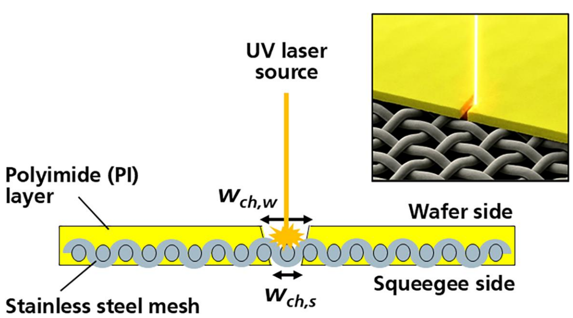

FIG. 7. Schematic of laser structuring process for fine-mesh screens with polyimide (PI) barrier layer. The contact finger openings (channels) are realized with a UV laser, which locally ablates the PI film from the wafer side. This results in a funnel-shaped (tapered) cross section of the channels, with the narrower opening width $w_{ch,s}$ on the squeegee side. Please note: image partly modified using artificial intelligence (AI) image generation.

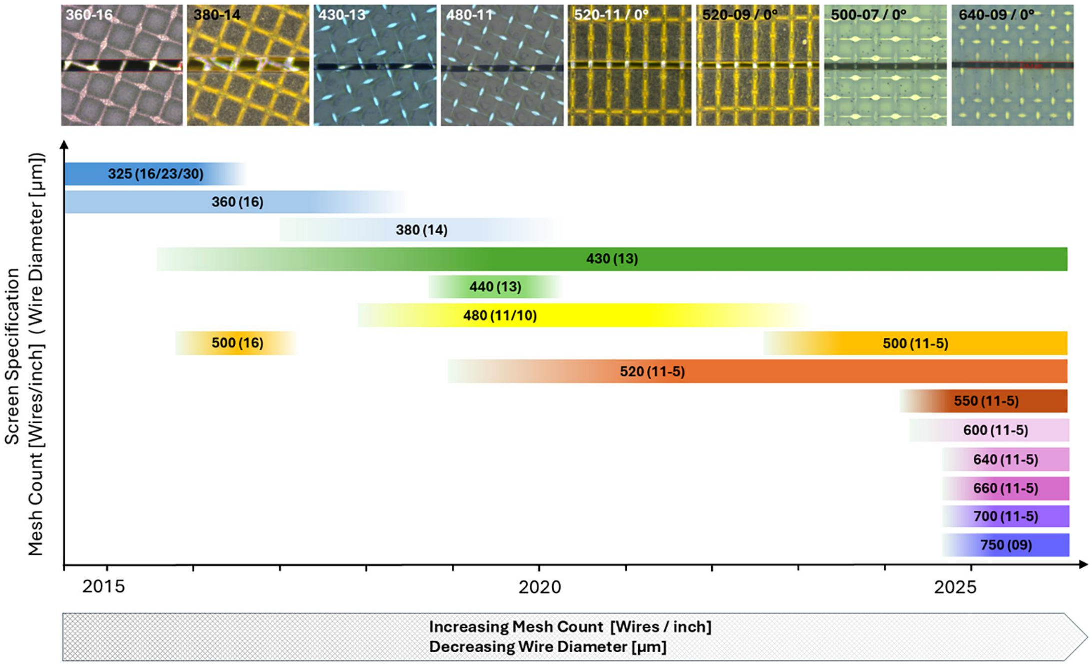

FIG. 8. Timeline showing the evolution of fine-mesh screen configurations from 2015 to today. It is clearly visible how the ongoing trend toward finer contacts with low-silver laydown accelerated the development of finer and finer screens with increasing mesh count and decreasing wire diameter. Due to the development of mesh back-etching, the bandwidth of different screen types increased dramatically from around 2023 onwards.

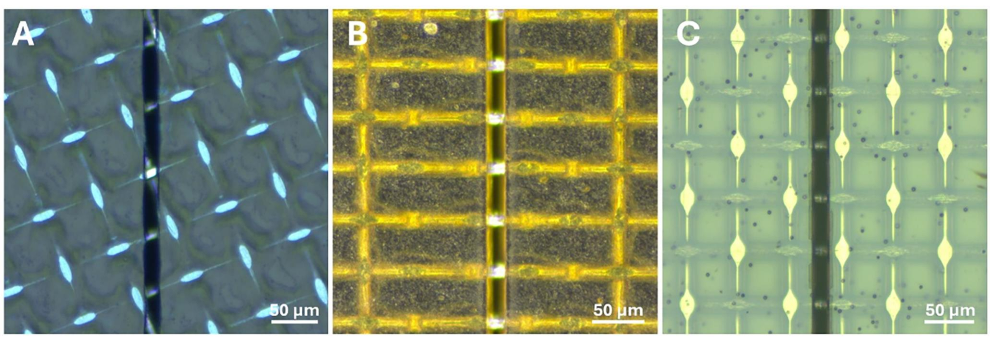

FIG. 9. Microscopic image of a conventional screen with 22.5° mesh angle (a) compared to a "knotless" or "zero-degree angled" screen with a mesh angle of 0° (b) and (c). In the case of knotless screens, the finger channels are lased in between two parallel wires which effectively reduces local deformations (mesh marks) by wire intersections in the channels. Originally, knotless screens have been fabricated by actively removing wires manually or via laser cutting (b). Over the past few years, the screen lasing process has been further optimized, enabling fine finger openings between the wires without prior wire removal and thereby preserving the original mesh stability during printing (c).

openings as a key parameter governing the resulting shape of the printed contacts. Higher slip velocity strongly correlates with improved paste transfer efficiency, reduces paste residue on the screen, and enhances electrode geometry including higher aspect ratios. Tepner *et al.*[178] introduced a "slip efficiency" parameter β, that combines critical slip stress and slip velocity formation time to predict printing performance.[178] Yet, the width of the printed contacts is assumed to be primarily controlled by yield stress rather than slip behavior.[177] The contact finger width primarily depends on the reciprocal of the product of yield stress and high shear viscosity.[179] Furthermore, it was proposed that the paste-wire interaction (work of adhesion at the paste-wire interface) fundamentally determines the print quality (shape of the contacts) and the print reproducibility.[177,180] Pushing screen printing toward ever-finer contact widths therefore requires careful optimization of relevant paste rheology parameters, mesh and screen properties, and printing conditions, which becomes increasingly challenging as production line throughputs—and thus printing velocities—continue to increase. Nowadays, fine-line metallization pastes are highly optimized for specific, standardized process conditions (screen or stencil configuration, printing conditions, emitter profile, passivation properties, wafer material surface, etc.) and continuously improved in very short iteration cycles. The cycle and printing speeds play an essential role in the printability of the finest lines. Typical industrial printing velocities are in the range of $v_{\text{print}} = 500\text{–}550$ mm/s, a flooding speed of $v_{\text{flood}} = 1000\text{–}1500$ mm/s, and a cycle time of around $t_c = 0.75$ s per M10 cell, which translates into a throughput of more than 9500 wafers per hour on the latest dual lane printing lines.[28]

Taken together, all these impact factors and developments led to a consistent and rapid evolution of the screen printing process resulting in a dramatic reduction of the average industrially printed finger width for solar cell metallization. This impressive effort has contributed significantly to the overall increase in the optical and electrical performance of modern solar cells and modules.

F. Modeling a learning curve for fine-line screen and stencil printing

The technological progress of screen printing for solar cell metallization can be easily traced on the basis of published results relating to the printed finger width as an essential quality factor for the performance which is also directly related to the silver consumption. In previous work, an extensive literature search of published screen printing results since 2008 was carried out and presented as a learning curve.[42,175] For this work, additional results published since 2023 were researched and added. It should be noted that this comprehensive collection of finger width results includes different cell concepts (Al BSF, PERC and TOPCon), as the predominant fabricated cell concept in industrial production changed multiple times throughout the last two decades by displacing older cell concepts (Al BSF) with new high-efficiency concepts (PERC and later on TOPCon). However, the aim of this study is exclusively to consider the technological development in screen and stencil printing for front-side metallization. To ensure comparability, the collected results refer exclusively to fine-line front-side metallization for high-temperature cell concepts using silver or silver-aluminum paste. Detailed information on the cell concept and specific paste type used can be found in the supplementary material (Table S1).

However, the following considerations and assumptions should be taken into account:

- Al BSF and PERC solar cells are almost comparable in terms of front-side metallization (contact fingers), as both cell concepts have an n^+-emitter, similar front-side passivation and largely comparable silver paste formulation.
- The n-TOPCon cell concept differs from these two cell concepts, as the front surface is equipped with a p^+-emitter. To contact this emitter properly, the front-side metallization of n-TOPCon cells has been originally realized with Ag/Al pastes. These Ag/Al pastes were limited with respect to fine-line printability due to the presence of larger Ag particles in the paste which led to clogging of very fine screen openings and fine meshes. This changed with the widespread introduction of LECO which enabled the use of pure Ag pastes optimized for fine-line printing.
- Apart from screen and paste properties, there are further influencing factors such as surface morphology (i.e., texture properties[181,182]) that have changed over time and can also have a minor effect on the printed finger width. However, this information is usually not documented in detail and therefore cannot be analyzed.

In addition, publications that report printing results often specify only individual finger widths obtained at the R&D level, without statistical substantiation such as production data. Nevertheless, in our view, the learning curve provides a good indication for the rapid progress in screen and stencil printing with respect to the printability of fine lines.

Figure 10 shows the updated linear learning curve, highlighting selected published screen- and stencil-printed finger width results from 2008 to 2025. The learning curve underlines the rapid progress in reducing front-side finger width from approximately 90–110 μm (2008) to $\sim$5 μm (on R&D level) in 2025. Selected scanning electron microscopy (SEM) images of representative contact fingers, which were printed at Fraunhofer ISE between 2013 and 2024, visually illustrate this impressive development (Fig. 11). These images also show that, in parallel with the reduction in width, the aspect ratio (height to width ratio) was increased and the shape of the contact fingers improved in terms of uniformity and flank steepness. The reasons for this impressive development were, on the one hand, technological advances in the field of screen production and structuring, and on the other hand, an increasing understanding of the rheological properties and flow behavior of the paste in the screen channel. This understanding led to a substantial optimization of formulations for high-performance metallization pastes.[177,178,180]

While the linear representation of the learning curve only shows the development of finger width as a function of time, it is also important to highlight its connection with the cumulative PV production

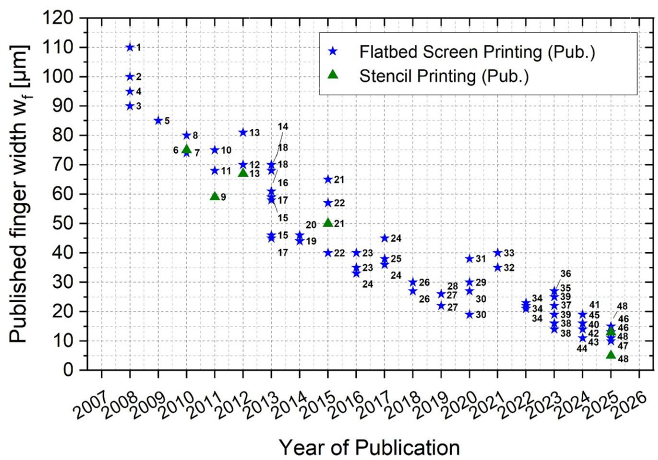

FIG. 10. Learning curve showing the reduction of screen- and stencil-printed finger width on solar cells (various cell concepts) between 2008 and 2024 based on published results. It is clearly visible that the finger width could be constantly reduced within this period. Adapted with permission from Tepner and Lorenz, Prog. Photovoltaics **31**(6), 557–590 (2023). Copyright 2023 Authors, licensed under a Creative Commons Attribution 4.0 International License (CC BY 4.0).[42] Literature references for this figure are provided in Table S1 in the supplementary material.

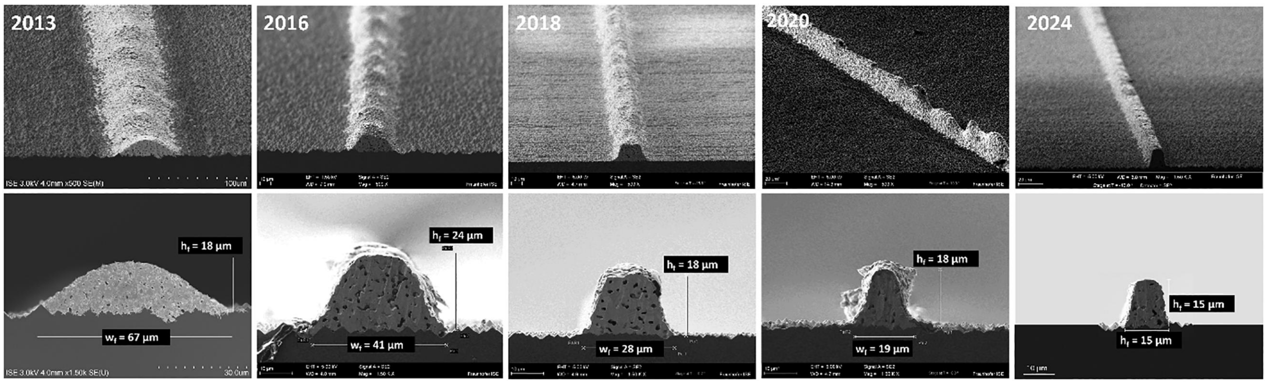

FIG. 11. Selected SEM images of Al BSF and PERC front-side contact fingers printed at Fraunhofer ISE from 2013 to 2024. It is clearly visible that the finger width was drastically reduced while in parallel increasing the aspect ratio. Furthermore, the shape of the cross-section is improved by printing steeper flanks. Overall, these improvements substantially helped to decrease shading losses, reduce silver consumption and increase the efficiency of silver usage by avoiding excessive silver paste usage in areas with low contribution to the lateral finger conductivity.

volume and with disruptive developments related to screen printing and cell interconnection. Thus, a two-phase learning curve based on published finger width data until 2024 has been modeled (Fig. 12). In addition, the globally produced growth in PV during this period was determined and combined with the data. For clarity, a logarithmic representation of the finger width was chosen. The learning curve was divided into two sections based on production capacity and fitted to the data with the following fit functions:

$$\text{Section 1 (Production capacity 10–100 GWp/a): } w = 151\,\mu\text{m} \cdot p^{-0{,}165}, \tag{1}$$

$$\text{Section 2 (Production capacity 101–2300 GWp/a): } 389\,\mu\text{m} \cdot p^{-0{,}369}, \tag{2}$$

with calculated finger width w and production capacity p.

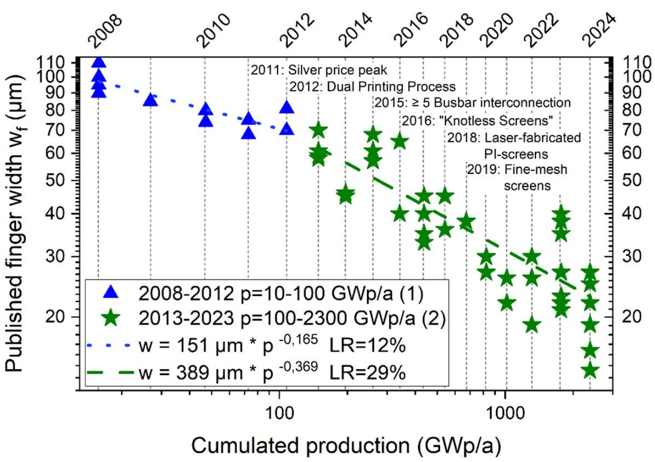

FIG. 12. Learning-curve model based on the same data as shown in Fig. 10 using a two-section fitting approach. Using the fitting functions (1) and (2), the curve shows a learning rate of 12% from 2008 to 2012 and a steeper learning rate of 29% from 2013 to 2024. Furthermore, important technological milestones are added based on Fig. 5.

With these assumptions, the learning rate is calculated with $LR_1 = 12\%$ for Section 1 (2008–2012) and $LR_2 = 29\%$ per doubling of cumulative capacity for Section 2 (2013–2024) using the fit functions (1) and (2), see Fig. 12. This results in a more differentiated picture of the development of finger width in screen printing. While the period from 2008 to 2012 was characterized by a moderate annual decrease in the average finger width, from 2013 onward there was a much steeper learning curve due to the technological developments already described resulting in an accelerated reduction of the average printed finger width. In particular, the transition from three to five or more busbars enabled a rapid reduction of the printed finger width and can thus be considered a "game changer" for solar cell metallization.

The newly modeled curve highlights a substantial acceleration of the technological learning associated with the screen printing process at a time, when the cumulated production volume exceeded 100 GW_p/a. This accelerated development was accompanied by various technological innovations in the field of screen printing and cell interconnection. In particular, the transition from 3 to 5 and more busbars set a decisive course for an even faster development of the fine-line screen printing metallization.

III. EVOLUTION OF SILVER CONSUMPTION FOR THE PV INDUSTRY

A. Validation of the silver sustainability limit of 2 mg/W_p

As discussed in the introduction, a long-term silver consumption target value of $t_{Ag} < 2$ mg/W_p was proposed by Zhang *et al.* in 2021.[7] Originally, the following assumptions were made for the calculation of t_{Ag}:

- A stable annual PV production of $P_a = 3$ TW/a to achieve a long-term target of 70 TW cumulative installed PV capacity in 2050[18]
- An estimated average total annual silver supply of $S_{Ag,tot} = 29\,000$ t/a in 2019
- A silver supply share of $f_{PV} = 0.2$ (20%) for the PV industry related to the annual total silver supply $S_{Ag,tot}$

The proposed limit of $t_{Ag} < 2\,mg/W_p$ has proven to be a very important guardrail for the PV industry, as it formulated a very specific goal for the sustainable use of silver as a critical resource supported the adoption of low-cost solar. By reducing the high dependence of PV on silver, increasing module prices due to sharply rising silver costs can also be avoided. However, the question arises whether this target can still be considered up to date and realistic in view of substantially changed boundary conditions today. Hallam *et al.*[5] calculated a silver learning rate of 20.3 ± 0.8% reduction in silver consumption for every doubling of cumulative PV capacity throughout the last decade. Assuming this learning rate and a total installed PV capacity of 63.4 TW in 2050, silver consumption would consequently fall to only 5.3 ± 0.5 mg/W_p, thus clearly missing the proposed target of < 2 mg/W_p. Provided that silver-intensive cell concepts like n-TOPCon and SHJ gain a strongly increasing market share (which is already the case today), the PV industry would theoretically require a cumulated silver supply of over 100% of all known global silver reserves with this scenario.[5] However, this also depends on the total amount of known global silver reserves, which were estimated at 610 kt according to the 2026 USGS Mineral Commodity Summaries.[183] To avoid this unrealistic scenario and thus a resource-restricted, premature slowdown of PV production, the authors propose a significant acceleration of the learning rate to 30%–40%. In 2025, Cattaneo *et al.*[8] projected a strongly increasing total silver demand of 48 000–54 000 t/a until 2030. The PV industry is expected to be the fastest-growing silver consumer with an expected demand of 9.794 t/a (conservative scenario) to 13 845 t/a (high-demand scenario).[8] The calculated scenarios show that the PV industry is actually moving toward a silver share of 30%–40% by 2030 and is thus moving further and further away from the original long-term target of $f_{PV} = 0.2$ postulated by Zhang *et al.* In parallel, competing sectors like electronics and automotive industry might account for an expected demand of 38 000–40 000 t/a until 2030.[8] Cattaneo *et al.* estimated a total silver supply of $S_{Ag,tot} = 33\,827$ t/a in 2030 by projecting the total historical silver supply since 2020 (without considering a possible expansion of silver mining due to a substantially increasing silver price).[8] Assuming this projected total supply would thus lead to a substantial supply gap of approximately 30%–38%. An increase in total silver supply, for instance through the expansion of mining activities or more efficient recycling processes,[184,185] would help to alleviate this constraint.

In order to check the validity and realistic feasibility of the previously proposed limit of $t_{Ag} = 2\,mg/W_p$, we calculate various scenarios based on the previous studies.[5,7,8]

We calculate the silver consumption limit t_{Ag} (mg/W_p) using the following equation:

$$t_{Ag}[mg/W_p] = (f_{PV} \cdot S_{Ag,tot}[t/a])/(1000 \cdot P_a[TW/a]). \tag{3}$$

The following assumptions are made:

- A deliberately conservative assumption of a constant total silver supply of $S_{Ag,tot} = 33\,827$ t/a from 2030 onward, based on Cattaneo *et al.*,[8] neglecting the possibility of any further increase from new mines or improved recycling.
- Silver supply share of the PV industry: $f_{PV} = 0.0$–0.5
- Annual PV production volume for a future terawatt-scale PV industry:
- $P_a = 1.0$–6.0 TW/a

Based on these assumptions, we perform a two-dimensional parameter sweep varying the parameters f_{PV} and P_a. Additionally, we calculate specific scenarios based on specific assumptions regarding the development of PV production volume and PV silver demand based on previous work.[5,8]

Scenario "Ref-2021" refers to the assumptions made by Zhang *et al.* for the calculation of the limit of < 2 mg/W_p based on a total silver supply of $S_{Ag,tot} = 29\,000$ t/a. As a comparison, an updated scenario "Ref-2025" is calculated based on latest estimates of the annual silver consumption of the PV industry $S_{PV} = 6087$ t/a,[11] the total silver supply predicted for 2025 with $S_{Ag,tot} = 32\,055$ t/a[11] and an assumed PV production volume of $P_a \sim 700$ GW,[2] resulting in a calculated actual PV silver share of $f_{PV} = 0.19$. Scenarios "2030-low," "2030-mid," and "2030-high" are based on previously calculated projections for a conservative to high-demand silver annual consumption in 2030 by Cattaneo *et al.*[8] Scenarios "2050-mid" calculates an updated silver consumption limit based on the assumptions of Zhang *et al.* ($f_{PV} = 0.2$, $P_a = 3$ TW/a). For simplicity, we assume that from 2030 onward total silver supply is capped at $S_{Ag,tot} = 33\,827$ t/y as projected by Cattaneo *et al.*[8] Scenario "2050-high" calculates the resulting PV silver consumption S_{PV} based on the calculated current learning rate of ∼20% and a resulting t_{Ag} ∼5.3 mg/W_p as projected for a "business as usual" case by Hallam *et al.*[5] Finally, a further ambitious scenario "2050-low" is calculated assuming a very low PV silver share of $f_{PV} = 0.1$ in 2050 considering a possibly strongly rising silver demand of other industry branches (i.e., automotive industry, artificial intelligence data center infrastructure, chemical industry, medical technology, as well as aerospace and defense[10]) Table I shows the selected scenarios, the relevant input parameters and the resulting calculated silver consumption limit t_{Ag}.

The results of the parameter sweep including the calculated, specific scenarios are visualized as a heatmap (Fig. 13).

Comparing the assumptions of Zhang *et al.* from 2021 ("Ref-2021") with the current, actual situation in 2025 ("Ref-2025"), the originally assumed long-term silver share of $f_{PV} = 0.2$ is already almost reached today (compared to a silver share of ∼10.3% at the time of the study[7]). Considering actual data regarding silver consumption and PV production volume results in a calculated average silver consumption of $t_{Ag} = 8.7\,mg/W_p$ for 2025. The strongly increasing silver consumption since 2021 is caused by an increasing production volume on the one hand and a strongly increasing market share of high-efficiency cell concepts (TOPCon, SHJ) on the other hand. This development confirms the significantly too slow real silver learning rate, as already indicated by Hallam *et al.*[5] Updating the assumptions of Zhang *et al.* with a moderately increased projected total silver supply of $S_{Ag,tot} = 33\,827$ t/a from 2030 onward[8] leads to an adapted, slightly increased silver consumption limit $t_{Ag} = 2.28\,mg/W_p$. Considering the currently projected scenarios of Cattaneo *et al.*[8] for a conservative, moderate, and high-volume silver consumption of the PV industry in 2030 results in a growing silver consumption t_{Ag} between 9.16 mg/W_p (for scenario "2030-low") and 10.65 mg/W_p (for scenarios "2030-mid" and "2030-high"), as well as a strongly growing silver share f_{PV} of 0.29–0.41. If the silver learning rate of ∼20.3% as calculated by Hallam *et al.*[5] is continued, the silver demand in 2050 would rise to about 15 900 t/a and thus a very high share of $f_{PV} = 0.46$, which would likely be incompatible with the increasing demand of other branches. This scenario is extremely unrealistic in

TABLE I. Calculation of the target silver consumption limit t_{Ag} for a sustainable multi-terawatt PV production for different scenarios with varying conditions using Eq. (3). A fixed total silver supply of $S_{Ag,tot} = 33\,827$ t/a based on the projection of Cattaneo *et al.*[8] from 2030 on is assumed for all scenarios apart from scenario "Ref-2021" with an assumed silver supply of $S_{Ag,tot} = 29\,000$ t/a and "Ref-2025" with current silver supply of $S_{Ag,tot} = 32\,055$ t/a.

Scenario	Annual PV production volume P_a (TW/a)	Annual PV silver consumption S_{PV} (t/a)	Silver share PV industry f_{PV} (-)	Average silver consumption t_{Ag} (mg/Wp)
Ref-2021 (Ref. 7)	3.00[a,b]	5 800	0.20[a,b]	1.93
Ref-2025	0.70[b]	6 087[b]	0.19	8.70
2030-Low	1.07[b,c]	9 794[b,c]	0.29	9.16
2030-Mid	1.07[b,c]	11 385[b,c]	0.34	10.65
2030-High	1.30[b,c]	13 845[b,c]	0.41	10.65
2050-Low	3.00[b]	3 250	0.10[b]	1.08
2050-Mid	3.00[b]	6 850	0.20[b]	2.28
2050-High	3.00[b]	15 900	0.46	5.30[b,d]

[a]Scenario Zhang *et al.*[7]
[b]Input parameters for calculation.
[c]Scenario Cattaneo *et al.*[8]
[d]Scenario Hallam *et al.*[5]

view of the rapidly growing competition from other industrial sectors with regard to silver demand. Since photovoltaics is a low-cost application today, the PV industry can hardly afford a very high cost share for silver. Rather, it is to be expected that the pressure for drastic and rapid silver reduction will continue to increase enormously due to sharply rising silver prices. If the learning curve cannot be accelerated quickly and drastically, there is a risk of a severe reduction in the possible future production volume due to sharply rising module prices and possible bottlenecks in the silver supply. This issue will become particularly relevant once future PV manufacturing actually reaches a scale in the multi-terawatt range.

These findings indicate a clear need for the PV industry to accelerate reductions in silver use and potentially adapt to a significantly lower silver supply share in view of future multi-terawatt-scale production. In view of the ever-increasing competition from different,

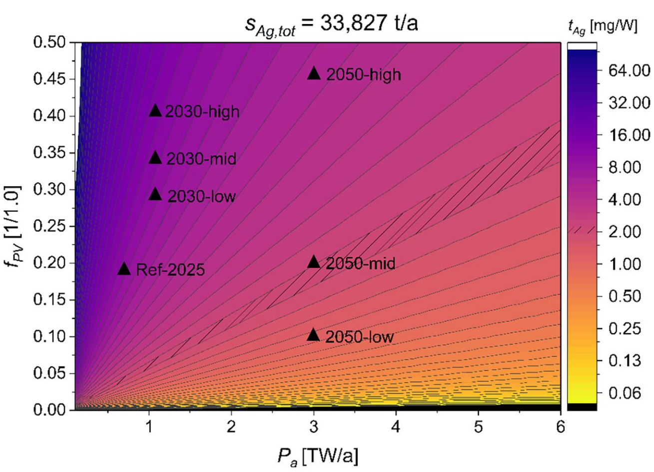

FIG. 13. Two-dimensional parameter sweep (heatmap) of t_{Ag} depending on f_{PV} and P_a for a total annual silver supply of $S_{Ag,tot} = 33\,827$ t/a including the different scenarios as shown in Table I.

fast-growing industrial sectors with regard to silver demand, the assumption of a long-term, sustainable supply share of 20% for the PV industry is increasingly hard to justify. The limit of $t_{Ag} = 2$ mg/W_p (or, updated to the future silver supply, $t_{Ag} = 2.28$ mg/W_p) postulated by Zhang *et al.* for an assumed annual production volume of 3 TW/a and a silver share of $f_{PV} = 0.2$ can thus continue to be regarded as an upper benchmark for future sustainable PV production. However, considering the actual development, we propose a more ambitious normative long-term sustainability goal of $t_{Ag} \sim 1$ mg/W_p (scenario "2050-low," $t_{Ag} = 1.08$ mg/W_p), which corresponds to a substantially lower silver share of only 10% ($f_{PV} = 0.1$) based on an assumed total annual silver supply of 33 827 t/a. However, it should be noted that this is a normative scenario with concrete specifications ($f_{PV} = 0.1$), which does not make any statement about the actual technical feasibility or changed framework conditions. Naturally, changes in future annual PV production volume or in total annual silver supply—for example, through new mining projects or more efficient recycling—will accordingly mitigate or exacerbate this scenario. It should be further noted that silver is predominantly mined as a by-product of copper, lead-zinc, and gold extraction.[186] The anticipated rapid growth in primary copper production[187] driven by electrification and low-carbon technologies could therefore partially alleviate silver supply constraints, although the extent depends on the silver content of newly developed copper deposits and ongoing ore grade decline.[188]

B. Factors influencing silver consumption in the PV industry

As already discussed in Sec. I, silver consumption in the PV industry continues to grow steadily and could reach 29%–41% of the annual total silver supply by 2030. This estimated strong increase in total silver consumption is of course based on the assumption of an ongoing strong increase in annual global PV production. At the same time, however, other factors should be considered, some of which contribute to an additional increase in silver consumption and some to a reduction in silver consumption. To assess the long-term progress of the PV industry in reducing silver consumption, it is most suitable to consider the average silver consumption in mg/W_p at the module level. However, calculating and comparing normalized silver consumption is not straightforward, as various factors have had a decisive influence on the actual silver consumption per cell over time. The following aspects must be considered:

- Silver consumption per cell has fallen sharply due to advances in metallization and interconnection (see Sec. II B), e.g., from about 300 mg per cell for Al-BSF in 2010 to about 74 mg per cell for PERC in 2023. However, these values are often estimated average values across a large number of manufacturers and individual cell designs, so the average silver consumption is only a rough approximation, for example as reported in the ITRPV.[14]
- Modern high-efficiency cell concepts such as TOPCon and SHJ, but also the latest generations of the PERC cell concept are based on a bifacial cell architecture with light coupling on both sides of the cell.[40] Compared with monofacial modules, bifacial designs capture rear-side irradiance from reflected and diffuse light, yielding higher energy production and improved performance, especially over high-albedo surfaces. However, bifacial cells require a highly conductive grid on both sides of the cell, which drastically increases the usage of silver paste compared to monofacial cell concepts.[189] The shift from monofacial Al BSF solar cells to monofacial (and partly bifacial) PERC solar cells around 2012–2013, and later to fully bifacial n-type TOPCon (and, to a lesser extent, SHJ) solar cells around 2022–2023, thus had a strong effect on the average silver consumption per cell,[189] temporarily offsetting the long-term reduction trend. The calculation of the average silver consumption per cell strongly depends on the assumed annual market share of the different cell concepts, which is based on estimated values and therefore also subject to a certain degree of uncertainty.
- The rapid evolution of even larger wafer formats from M0 (156 mm × 156 mm) to M10 (182 mm × 182 mm), M12/G12 (210 mm × 210 mm), and even larger formats substantially increased silver consumption per cell.[14] At the same time, the average efficiency and thus the average maximal power output of silicon solar cells and modules has been constantly increased by introducing new high-efficiency cell concepts and numerous technological improvements along the whole process chain. Both parameters are essential input parameters for the calculation of the mean, normalized silver consumption in mg/W_p and are therefore also subject to a certain uncertainty of the input parameters.
- Silver pastes consist of only a certain proportion (usually 90–95 wt. %) of pure silver particles, while the remaining components include glass frit, solvents, binders, and additives that enable the required printability and contact formation of the paste. The actual silver content is slightly different for each paste, so that the assumed values for silver consumption are an approximation.
- The material substitution of silver by other materials (e.g., silver-coated copper pastes, hereinafter referred to as AgCu) is already widely used, especially for SHJ solar cells.[139,190] To a certain extent, this counteracts the increasing consumption of silver. For an accurate calculation of silver consumption per cell, the proportion and composition of AgCu pastes would therefore have to be taken into account. However, this data are not available, so the actual silver consumption is likely to be lower, especially for SHJ solar cells and modules. With respect to TOPCon, the front- and rear-side metallization is still realized primarily with silver pastes. However, this might change in the near future due to new metallization concepts which are already adopted in the industry[191] (see Sec. IV C).

C. Learning curve of silver's module cost share

Table II shows the evolution of the average annual silver price (USD/Kg), the assumed market share and average consumption of silver per cell for the most relevant cell concepts from 2021 to 2024 based on yearly ITRPV reports.[14,192–194] Based on these data, an average silver consumption per M10 sized cell considering the yearly market share of all relevant cell concepts has been calculated. The average silver consumption is further converted into milligrams per W_p at the module level considering a 60 cell M10 module and the average module power from the ITRPV reports. The data show that the average silver consumption per cell initially decreased from 2021 to 2023 due to technical advances in the metallization process but has been increasing substantially again since 2024. The increase originates primarily from the strong increase in market share for high-efficiency cell concepts (especially n-TOPCon), which have a significantly higher silver consumption than (monofacial) PERC solar cells due to bifacial designs requiring conductive grids on both sides.

TABLE II. Evolution of the average annual silver price, market share of relevant cell technologies and mean silver consumption per cell concept, as well as the calculated average silver consumption based on data from the annual ITRPV reports from 2022 to 2024.[14,193,194]

Parameter		2021	2022	2023	2024
Avg. annual silver price (USD/Kg)		81 013	70 732	73 947	90 802
Market share (%)	p-type PERC	81	80	64	35
	n-type TOPCon	4	10	29	56
	SHJ	5	8	5	9
Silver consumption per cell (M10) (mg/cell)	p-type PERC	100	80	74	70
	n-type TOPCon	145	145	120	110
	SHJ	210	200	160	140
Average silver consumption (all cell concepts) (mg/cell)		**97.3**	**94.5**	**90.2**	**98.7**
Average module power (60 cells per module) (W_p)		**403**	**409**	**426**	**451**
Average silver consumption (all cell concepts) (mg/W_p)		**14.5**	**13.9**	**12.7**	**13.1**

For the year 2024, we calculate the remaining amount of silver as an average across all cell technologies (PERC, TOPCon, SHJ) weighed by market share, which results in 98.7 mg/cell for an M10 sized cell. Considering an average module power of 451 W_p for a 60 cell module (full size cells) with M10 cell format in 2024,[14] the average remaining silver consumption can be calculated with approximately 13.1 mg/W_p. We combine the rise of the annual average silver price as of 2024 (see Fig. 1) with the year-over-year reduction in silver consumption per cell and module. Assuming an M10 module with 120 half-cut cells (the dominant 2024 module concept for PERC and TOPCon), we then calculate the share of silver in the module price in USD/W_p.

The yearly share of silver in PV modules S_{Ag} in USD/W_p is calculated using Eq. (4), based on the remaining silver per cell $m_{Ag,cell}$ (in mg/cell) and the number of cells per module n (60 full cell or 120 half-cut-cell modules assumed for the period 2010–2024). This remaining silver per module (in g/module) is then multiplied by the prevailing silver metal price p (in USD/g) for the share of silver price in the module (in USD/module). Dividing the share of silver price with the corresponding module power P_{mod} (W_p) in that year, we get the share of silver price S_{Ag} in USD/W_p,

$$\mathrm{S_{Ag}[USD/W_p] = m_{Ag,cell}[mg/cell] \cdot n\,[cells]/1000\,[g/mg] \cdot \left(p\,[USD/g]/P_{mod}[W_p]\right).} \tag{4}$$

With the remaining silver per cell $m_{Ag,cell}$ (mg/cell), the number of cells per module n, the silver price p (USD/g), and the module power P_{mod} (W_p).

Figure 14 shows the PV module learning curve to visualize the learning rate (LR) with the cumulative PV production. The LR is calculated based on the following formulas:[195]

$$b = -LOG\left(\frac{c_t}{c_0}, \frac{q_t}{q_0}\right), \tag{5}$$

$$LR = 1 - 2^{-b}, \tag{6}$$

where b is a constant and refers to the learning by doing elasticity, which shows the percentage change in cost as a result of one percentage point rise in cumulative capacity. c_t and c_0 refer to the module price in years t and 0, and q_t and q_0 refer to the cumulative global production in years t and 0. The learning rate reveals by what percentage the cost decreases when production is doubled and is calculated from the learning by doing elasticity constant b.

Table III summarizes the key input data and the associated calculated LRs, which are shown in Fig. 14.

The PV module learning curve LR_{mod} shows the observed reduction in the price of PV modules with every doubling in PV production. The historic LR of PV modules starting from the year 1980 to 2024 is calculated as 26.2% based on Eqs. (5) and (6). The year 2010 can be considered as the start of PV mass production with a strongly increasing production capacity in the following years with a strongly increased learning rate of $LR_{Mod} = 38.6\%$, implying a corresponding PV module price reduction for every doubling in global PV production capacity during this period. At the end of 2024, global cumulative c-Si PV production finally reached the impressive amount of 2.34 TW_p at a median module price of 0.13 USD/W_p.[21] This corresponds to a 94% reduction of the PV module price learning curve LR_{mod} from 2010 to 2024.

The PV efficiency LR_{Eta} of 7.1% might seem comparatively low at first glance. However, this result shows that efficiency improvement alone resulted in a 35% reduction in the PV module price within the considered period (2010–2024). This means that efficiency

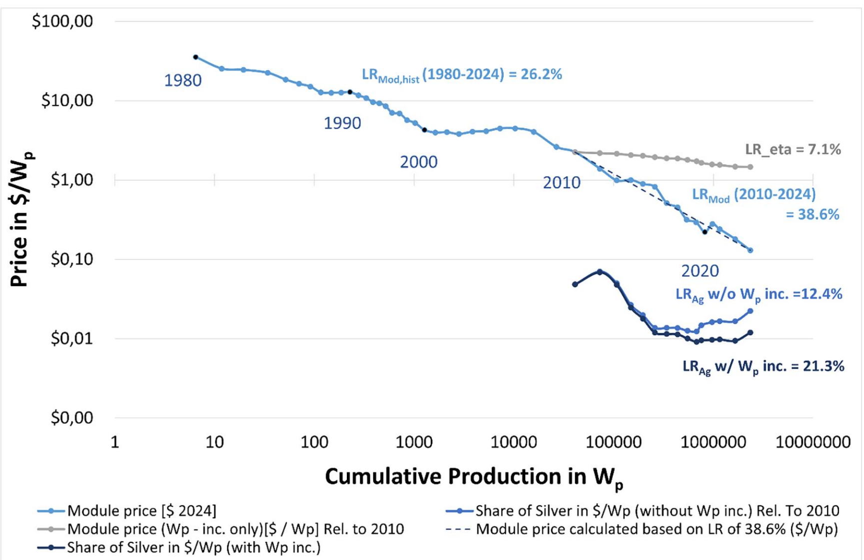

FIG. 14. PV module learning curve LR_{mod} plotted for module price in USD/W_p linked to the global cumulative production in W_p from 1980 to 2024. The historic learning curve $LR_{mod,hist}$ from 1980 to 2024 implies a learning rate of LR_{Mod} of 26.2%, while the learning curve since 2010 has a substantially increased learning rate of $LR_{Mod} = 38.6$. The LR_{Eta} due to efficiency improvement only is 7.1% from 2010 to 2024. The LR_{Ag} of the silver share in PV modules, relative to the year 2010 is 21.7% (considering the module efficiency increase) and 12.4% (neglecting the module efficiency increase). The dotted vertical line signifies the PV module price (0.13 USD/W_p) and cumulative PV production (2.35 TW_p) achieved by end of 2024.

TABLE III. Key parameters including the percentage change for calculating the learning rates for 2010 and 2024 as well as the calculated learning rates.

Year	2010	2024	Absolute change	Relative change	LR
Cumulative production (GW)	41.1	2348	2307	5607%	...
Module price (USD/W_p)	2.24	0.13	−2.11	−94%	38.6%
Module price W_p inc. only (USD/W_p)	2.24	1.46	−0.78	−35%	7.1%
S_{Ag} w/o mod. power increase (USD/W_p)	0.050	0.022	−0.038	−76%	12.4%
S_{Ag} with mod. power increase (USD/W_p)	0.048	0.012	−0.036	−75%	21.3%

improvements alone, translated into an increase in module power, reduced the price of PV modules by 7.1% with every doubling in cumulative PV production. Similarly, the ongoing reduction in silver consumption in PV modules resulted in a 75% drop in S_{Ag} when considering module power increases, and 54% when not. If the module power increase is neglected, the learning rate LR_{Ag} is still 12.4% meaning that the silver consumption (excluding the impact of any other factors like efficiency, productivity increase, etc.) decreased by 12.4% for every doubling in cumulative PV production from 2010 to 2024. Considering the constant increase in module power, a silver share learning rate $LR_{Ag,PMod}$ of 21.3% is determined. The higher learning rate results from the combination of an increase in module power due to the constantly increasing module efficiency, as well as the simultaneous reduction in the silver paste consumption. Particular attention should be paid to the uptick in S_{Ag} and its learning-curve metrics from 2023 to 2024. This uptick reflects higher per-cell silver use (TOPCon share increase) and higher silver prices as shown in Table II. Comparing the here calculated silver share learning rate of 21.3% with a previous study by Hallam *et al.*,[5] which had a learning rate of 20.3 ± 0.8%, the results are found to be in close agreement with each other. The difference in the current work is the separation of the learning rate of the silver consumption both with ($LR_{Ag,PMod}$ of 21.3%) and without (LR_{Ag} of 12.4%) an increase module power whereas that of Hallam *et al.* only includes the silver consumption learning rate with the increase in module power.

In summary, the results show that the rapid transition to high-efficiency cell concepts (TOPCon/SHJ), combined with a sharply rising silver price, is greatly increasing the economic pressure on the PV industry to rapidly reduce silver consumption.

IV. PATHWAYS TOWARD A SUSTAINABLE PV PRODUCTION

Given the predicted continued strong increase in annual PV production volume, the increasingly critical situation regarding silver supply—and thus rising costs for metallization and interconnection—is intensifying the pressure for a rapid and sustainable reduction in silver consumption. In the following, we discuss technological approaches and relevant developments to effectively reduce silver usage.

A. Silver reduction perspectives for screen printing

Currently, the fast development of new screen configurations with ever-finer meshes continues unabated. An overview of current screen and stencil configurations is shown in Table IV.

The latest knotless fine-mesh screens are fabricated with a mesh count of 700–760 wires per in. and a (back-etched) wire diameter of only 4–5 μm.[37] Newly developed mixed woven mesh screens are fabricated with different mesh counts (e.g., 300-08/500-11 mesh) in the warp and weft directions, which increases the open area and thus improves the uniformity of the printed contacts.[139] A relatively new approach is the nickel-alloyed pseudo-mesh screen, which is fabricated using a nanoimprint and subsequent electroplating process.[37] This approach enables increased mechanical strength and extended lifetime, excellent print quality for very fine contacts, and a 20%–30% cost reduction compared with conventionally woven mesh screens.[37] Furthermore, this approach avoids local thickening at wire intersections caused by mesh calendering,[37] whereby the pseudo-mesh screen has a lower roughness of the PI-coated surface, which presumably supports the edge sealing of the channels. There is also continued progress regarding the optimization of the laser structuring process for PI screens. R&D results obtained in 2025 show that finger openings down to 1 μm in the PI barrier layer can be realized on fine-mesh screens and stencils.[196–198] Current fine-mesh screens enable stable and reproducible printing of contacts down to ∼10 μm printed finger width.[37,139] Realizing even finer contacts in industrial mass production using mesh-based screens is questionable. This would probably require even finer meshes than today's most advanced configurations, with very thin wires, even thinner PI films, and rheologically highly optimized nanoparticle pastes. Latest R&D results[37] show that it is probably more promising to use metal-foil stencils when aiming for very fine contact fingers in the range of ∼5 μm printed finger width or less.[37]

B. Renaissance of stencil printing—a promising alternative to mesh-based screens

The increasing costs of fine-mesh screens, as well as emerging technological limits for mesh-based screens, are currently leading to a renaissance of stencil printing as an alternative to screen printing. Stencil printing already has a decades-long history in the field of solar cell metallization. Inspired by printed circuit board manufacturing, ECN began evaluating the stencil printing process for printing front-side contacts for silicon solar cells back in the mid-1990s.[199,200] The aim was to use metal-foil stencils as an alternative to mesh-based screens in order to print contact fingers with an increased aspect ratio and a more uniform finger geometry, without disturbing mesh marks, thereby reducing both series resistance losses and optical losses due to the grid.[199] Stencils for the electronics and PV industries are typically fabricated by either direct laser structuring of a thin nickel or stainless steel foil[199] or using nanoimprint lithography[201] with subsequent electroforming.[37,202]

A specific challenge for stencil printing on silicon solar cells is the insufficient lateral sealing of the fine finger channels on the textured wafer surface. When printing with a single-layer stencil, the paste is pushed into the narrow gap between the bottom of the stencil

TABLE IV. Evolution of screen and stencil configurations, including estimated typical screen lifetimes (printing cycles) taken from Ref. 139, relevant properties, and economic effects of the various approaches.

Approx. year of industrial implementation	Screen type	Mesh types (wpi/μm)	Mask layer	∅ Screen Lifetime	Properties	Cost-effect
Until 2019/2020	Emulsion, 22.5°	290/20–480/11	Photoemulsion	~30–50k	Standard screen, limited resolution	Reference
~ 2018	PI, knotless, selective wire removal, no back-etch	480/11–640/09	Polyimide (laser-structured)	~100k	High resolution, increased lifetime	Silver reduction, Reduced screen costs, uptime
~2022	PI, knotless, no wire removal, back-etched	500/09–760/04	Polyimide (laser-structured)	~450–800k	Higher resolution, considerably increased lifetime	Silver reduction, Reduced screen costs, uptime
~2025	PI, mixed woven mesh	i.e., 300–08/500–08	Polyimide (laser-structured)	~100–450k	Increased open area/ finger uniformity	Silver reduction
~2025	PI, nickel-alloyed "pseudo-mesh"	600/06–760/05	Polyimide (laser-structured)	~450k	Flat intersection, high density	~20%–30% cost reduction
~2025/26	Metal-foil stencil		Polyimide (laser-structured)	>500 k	Higher resolution, increased lifetime and finger uniformity	Silver reduction, reduced stencil costs, uptime

and the textured wafer surface along the channel edges. This leads to excessive spreading of the printed contacts and thus to additional shading and silver consumption,[42] as shown in Fig. 15(a).

This effect intensifies with an increasing roughness (pyramid size) of the texture.[181,203] When mesh-based screens are used, the flexible barrier layer (photopolymer or PI) effectively prevents this effect by sealing the channel edges against the texture—an effect commonly known as "gasketing" or "sealing effect."[203] A similar solution for stencil printing is the use of hybrid or dual-layer stencils with an additional flexible barrier layer on the metal foil, as shown in Figs. 15(a) and 16. Two-layer stencils are long established, e.g., for solder bump printing.[204] The barrier layer enables a significantly better edge sealing of the channels, and on the other hand, it also improves the mechanical stability of the stencil.

The grid design of typical solar cells with numerous long, parallel contact fingers poses a particular challenge for the design and production of stencils. To ensure the stability of the stencil during handling and the printing process, linear elements must be kept short or stabilized by regular support structures in the channels (bridge elements). However, these bridge elements can lead to interruptions of printed contact structures, particularly when using a single-layer stencil. Another challenge is posed by pad-like structures, orthogonal structures (e.g., redundancy lines), and busbar elements in the grid layout. Stability problems of such structures can be solved by a special design of mesh or bridge-like support elements.[28] Usually, a dual printing process using two stencils or a combination of stencil and screen is applied to print finger and busbar elements separately.

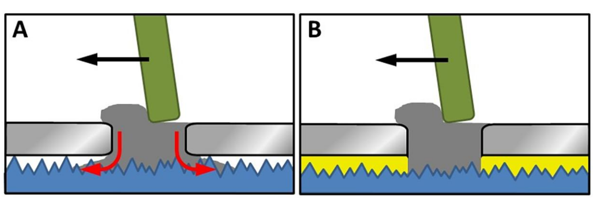

FIG. 15. Schematic of a stencil printing process on textured silicon surface without (a) and with (b) a flexible barrier layer (i.e., Polyimide). While the paste spreads excessively between the stencil underside and the wafer surface due to insufficient sealing when using a single-layer stencil, this is effectively prevented by the sealing function of the barrier layer in a dual-layer stencil.

As of 2025, major tier-1 manufacturers have started to use stencils in mass production of SHJ[139] and TOPCon solar cells.[37,139,202,205] According to the assessment of an industrial expert, the main advantage is the improved uniformity of the contacts compared with screen printing,[139] resulting in an efficiency gain of around 0.03%–0.05% or a silver reduction of 2–3 mg at the same efficiency level.[202] Yet, stencil printing also has promising potential for front-side metallization, as narrow fingers with widths in the range of 10–12 μm and aspect ratios above 0.7 can be obtained.[139,202] Latest R&D results have even demonstrated uniform stencil-printed fingers down to a width of ~5 μm.[37] However, stencil printing of such fine lines requires specific pastes with high thixotropy and comparatively high viscosity. These issues currently pose significant challenges for process stability in mass production according to the opinion of industry sources.[202]

C. Silver-lean metallization using alternative materials

A material-based approach to effectively reduce or even completely avoid the use of silver is the substitution of conventional

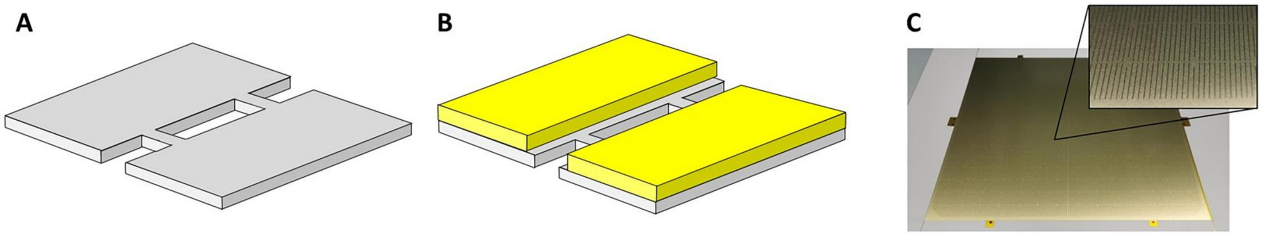

FIG. 16. Schematic of a single-layer (a) and dual-layer stencil with an additional polyimide sealing layer (yellow) (b). Digital image of a metal-foil dual-layer stencil for solar cell metallization. (c) Photography of a polyimide-coated stainless steel stencil demonstrating the homogeneous PI film coverage on the metal substrate.

silver pastes with other materials. The most promising alternative material is copper, as it has a specific resistivity similar to silver at substantially lower cost.[206] A promising approach is the partial replacement of silver particles in the paste with cheaper silver-coated copper particles.[207,208] Such pastes, hereafter referred to as AgCu paste, are already widely used for the industrial-scale metallization of SHJ solar cells.[209] AgCu pastes consist of a composite metal powder of larger copper particles coated with a thin layer of silver to protect against oxidation. Additionally, a certain proportion of silver nanoparticles can be added to increase packing density, sintering behavior, specific conductivity, and oxidation resistance,[208,210] as shown in Fig. 18(a). The core–shell structure of the AgCu particles effectively prevents oxidation at typical curing temperatures for SHJ solar cells of around 200–250 °C, provided that the silver shell has a sufficient thickness.[208,211] Further studies investigated the oxidation stability, which depends primarily on the sintering temperature as well as the thickness and density of the Ag shell.[212,213] Replacing silver pastes with AgCu pastes offers a potential to reduce the total silver consumption by up to 70%–80%, as demonstrated in 2025.[208,209] The latest AgCu paste generations have a remaining silver content of only 15%–20% (passing standard International Electrotechnical Commission (IEC) reliability tests) and even down to 10%–15% at the R&D level.[209] Dong *et al.*[208] and Tang *et al.*[214] showed that AgCu pastes provide comparable or even better conductivity than pure silver pastes at significantly reduced silver consumption. Using AgCu pastes, industrial manufacturers have already reduced the silver consumption of mass-produced SHJ modules to 4.5 mg/W$_p$, with a projected further reduction to 4.0 mg/W$_p$ within this year by introducing 0BB interconnection technology.[139] Using AgCu paste with a silver content of only 10% and photon-induced curing (also known as "intense pulsed light"[215]) might reduce silver consumption to as low as 3.5 mg/W$_p$ in mass production according to a 2025 estimate.[139] At the R&D level, a reduction of the total silver laydown to 1.4 mg/W$_p$ per SHJ cell has already been demonstrated using AgCu and pure copper pastes for front- and rear-side metallization.[23] Mini modules fabricated with these cells showed a promisingly low degradation of less than 2%$_{rel}$ after an accelerated thermocycling (aTC) test. However, damp heat (DH) tests still need to be conducted to confirm module reliability according to the IEC 61215 standard.[211]

A remaining challenge is the optimization of AgCu pastes for fine-line screen printing. The presence of larger AgCu particles in such pastes can lead to local clogging of fine-mesh screens with narrow finger channels, thus resulting in finger interruptions and constrictions.[211] Ag nanoparticles added to increase the packing density can lead to pronounced spreading of the contact fingers and thus to additional shading losses if the edge sealing between the textured wafer surface and the PI layer of the screen is insufficient [Fig. 17(b)]. Reducing the size of the AgCu particles is challenging due to the increasing oxidation tendency of smaller copper particles (e.g., caused by their higher specific surface area and other influencing factors[216]). AgCu pastes based on nanoparticles might be an option to overcome this limitation and enable the use of such pastes on the front side, provided that the particles can be fabricated at reasonable cost. From a technical point of view, the feasibility of fabricating screen-printable

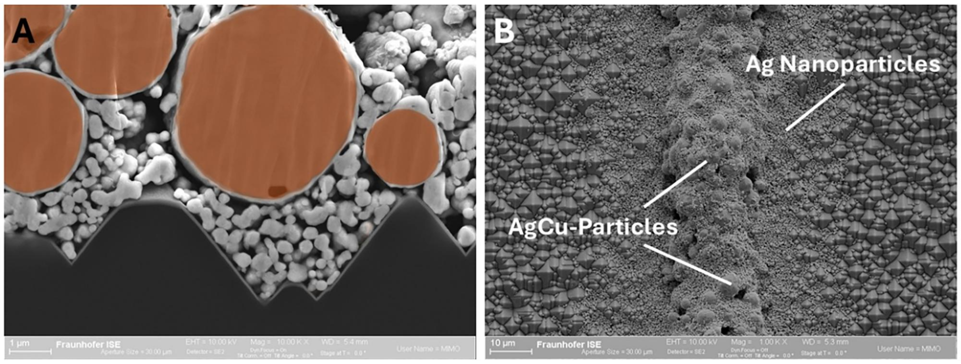

FIG. 17. (a) Colored SEM image of a AgCu paste containing copper particles coated with a thin silver layer to protect the copper from oxidation. (b) SEM image of a printed and cured contact using low-temperature AgCu paste. Clearly visible is a core finger zone of larger AgCu particles and a boundary zone consisting of nanometer-sized Ag particles, which increases the shading-relevant finger width due to excessive spreading on the textured wafer surface.

AgCu nanoparticles has been successfully demonstrated for various purpose, including power-electronic applications[217] and high-performance joints.[218] In 2022, Deng and Subramanian[219] demonstrated a silver reduction potential of 36 wt. % for solar cell metallization using AgCu nanoparticle pastes while achieving conductivity and optical reflection comparable to commercial silver pastes. However, AgCu nanoparticle pastes can suffer oxidation of the copper core during sintering in air at temperatures below 200 °C, as dewetting of the thin silver shell exposes the copper and leads to progressively stronger oxidation with increasing temperature.[218] In addition to AgCu nanoparticle pastes, metallization pastes based on silver nanoparticles could also be an interesting option for the realization of ultra-fine front contacts with reduced silver consumption. Compared to conventional pastes based on micrometer-sized particles, silver nanoparticle inks or pastes can achieve high specific conductivity at lower process temperatures and offer improved fine-line printability.[220,221] These results suggest that dedicated silver nanoparticle pastes are a promising option for ultra-fine SHJ front contacts with efficient silver utilization, provided that their material and processing costs remain competitive with current low-temperature Ag pastes.

While already widely established in the metallization of SHJ solar cells, the use of AgCu paste can also be a promising approach to reduce silver consumption for TOPCon solar cells.[15] Due to the sharply rising silver price, a large-scale replacement of silver pastes with copper-containing pastes in industrial PV production is expected in the short term.[191,222] A promising option for n-TOPCon solar cells is a dual-printing scheme for rear-side metallization using AgCu paste[189] or pure Cu paste,[223] as illustrated in Fig. 18. In the first step, a thin silver seed layer is printed on the rear side of the cell using a high-temperature contacting silver paste. After contact firing, a conducting grid is precisely printed onto the seed layer grid using low-temperature AgCu paste. Combined with ultra-fine-line metallization on the front-side p^+ emitter, this approach can help to effectively reduce overall silver usage for TOPCon.[37,139] Zhang *et al.* showed a 15% silver reduction on TOPCon cells using a screen-printed silver seed layer with a subsequent dual-printed AgCu conducting layer (20% silver share), and they also confirmed module reliability by passing a 2000 h damp-heat test.[139] In 2025, a silver reduction of ∼40% with a total silver consumption of only 7 mg/W_p was demonstrated on n-TOPCon solar cells by UNSW.[15] This was achieved using intermittent Ag dashes for contact formation and floating silver-lean or silver-free fingers and busbars for electrical conduction. Lossen *et al.* demonstrated a similar approach using a snap-cured copper paste as the conducting layer, proposing a silver reduction potential down to only 9 mg/W_p.[137] In 2025, a further promising approach was demonstrated using rear-side LCO and screen printing with AgCu paste, resulting in a 40%–60% silver reduction on the rear side without deteriorating the performance of the cells.[224]

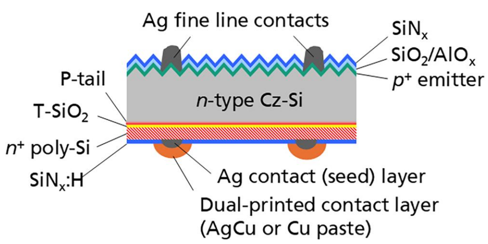

FIG. 18. Schematic of n-TOPCon cell with rear-side contacts consisting of a silver seed layer and a dual-printed AgCu or copper conducting layer.

While AgCu pastes provide an effective path to drastically reduce silver consumption on both SHJ and TOPCon solar cells, the use of pure copper pastes might be a long-term alternative to completely avoid silver usage. As of 2025, promising results on interdigitated back-contact (IBC) solar cells and SHJ solar cells using a snap-cured copper paste have been demonstrated.[225] While replacing silver pastes with AgCu and copper pastes is comparatively easy due to the relatively low curing temperature of around 200–250 °C, it is much more challenging to replace silver with pure copper pastes, particularly for high-temperature cell concepts like TOPCon. When fired at high temperature, copper rapidly penetrates and diffuses into silicon, forming deep-level traps and precipitates under certain conditions, which introduce defects and band-like states that negatively impact solar cell performance and carrier lifetime.[226,227] Furthermore, copper readily oxidizes during thermal treatment depending on the temperature and ambient atmosphere.[228] Further challenges arise with respect to cell interconnection (soldering) and long-term reliability of copper electrodes within the module.[229] Copper pastes pose the risk of copper cross-contamination throughout the production chain of solar cells. However, very promising results have been achieved on PERC and TOPCon solar cells using an innovative firing-through copper paste that effectively prevents oxidation and in-diffusion of copper through the passivation layer.[228,230–232] In 2025, TOPCon solar cells with a maximum efficiency of ∼24.4% have been demonstrated with this approach.[233]

Another material-based approach to substituting silver is the use of aluminum pastes for the backside metallization of n-TOPCon solar cells. Ourinson *et al.* demonstrated an approach using full-area screen-printed Al metallization on n-TOPCon solar cells with LCO, resulting in a 0.9% efficiency gap due to paste-related damage on the rear-side poly-Si layer.[223] Cheng *et al.* investigated the potential of screen-printed rear-side metallization using a specifically adapted Al paste containing Al-Si particles and tailored glass frits for n-TOPCon cells.[234] While the study successfully demonstrated the general feasibility of this approach with an efficiency gap of 0.8% to silver-metalized reference cells, it also revealed the necessity of further research and development with a focus on metal-related recombination, firing window tolerance, contact resistance and paste chemistry.[234]

Unsur *et al.*[22,235] proposed a promising approach based on replacing silver paste with a screen-printed silver-doped (4 wt. %) nickel paste with a tailored glass-frit composition. During the firing step, the glass-frit etches the SiN_x anti-reflection layer and then solidifies into a thin hole-selective glass layer between metal and emitter, providing a passivating, low-resistance contact.[22] This approach enables a drastic silver reduction potential of ∼96% leading to a minimal silver consumption of only ∼0.44 mg/W_p. While a closely comparable cell efficiency on n-TOPCon could be successfully demonstrated, further R&D will be necessary with respect to optimized glass frits, contact adhesion, robust industrial process windows and demonstration of long-term module reliability.[22]

Considering the increasing shortage of silver, it is likely that copper and other materials such as nickel will play an increasingly important role in the metallization of solar cells. In the short term, the use of screen-printed AgCu pastes is particularly promising, as silver consumption can be drastically reduced while avoiding serious problems associated with pure copper pastes.

V. SUMMARY AND CONCLUSION

This review sheds light on the technological development of screen- and stencil-printing processes for solar cell metallization over the past 40 years. A two-phase learning-rate model links the reduction of printed finger width—from ∼100 μm in 2008 to ∼10–15 μm in production and ∼5 μm at the R&D level today—to cumulative installed PV capacity, with an acceleration of the learning rate from 12% to 29% after 2013, which we trace back to economic constraints, increasing pressure to reduce silver consumption, and rapid shifts in interconnection concepts. This acceleration has been made possible by disruptive technological innovations and continuous development in the field of screen-printing metallization. Particularly noteworthy are the development and rapid spread of laser-structured, polyimide-coated screens in 2018, the rapid development of wire-weaving and back-etching technology for ultra-fine meshes, the introduction of the LECO process, and the development of knotless screens to realize very fine, high-quality contacts. At the same time, enormous efforts were made to better understand the flow properties of the metallization paste and to optimize paste formulations for fine-line printing. Advances reported in 2025 in high-performance stencil printing have improved finger shape and aspect ratio, enabling ultra-narrow contacts down to ∼5 μm while boosting the efficiency of silver use. Thus, we expect that stencil printing will gain widespread importance in solar cell metallization, as mesh-based screens may reach limitations regarding ultra-fine-line printing.

Using a two-dimensional parameter study of the relevant influencing factors, we evaluate the validity of the proposed limit of 2 mg silver per W_p for long-term sustainable multi-terawatt-scale PV production. We conclude that even this ambitious limit is probably set too high under current boundary conditions and propose a lower limit of ∼1.0 mg/W_p. The current estimated average silver consumption of ∼8.7 mg/W_p for 2025 highlights the substantial effort required to achieve this long-term goal.

We further investigate the module-price learning rate and obtain a value of 38.6% per capacity doubling from 2010 to 2024. Based on this result, we calculate a learning rate of 21.7% per doubling for silver's cost share, normalized to module power. The learning curve also shows a recent uptick, which we trace back to the increasing market share of TOPCon solar cells and a sharply rising silver price.

Finally, we discuss technological and material options for a sustainable and substantial reduction in silver consumption for solar cell metallization. In addition to printing ever-finer contacts, material substitution offers major potential for silver reduction. AgCu pastes are already widely used for SHJ cells and are promising for TOPCon rear-side metallization, enabling drastic reductions in silver use without sacrificing performance, whereas pure copper paste represents a longer-term option that still face significant challenges regarding diffusion, oxidation, and long-term reliability. Further approaches based on silver-doped nickel and aluminum pastes offer additional promising routes to reduce, or even completely replace, silver in the long term.

The remarkable adaptability of screen and stencil printing, together with continuous innovation in screens and stencils, paste composition, and machinery, indicates that screen and stencil printing are well positioned for future solar cell metallization, even in an era of material constraints and terawatt-scale production requirements. From a technical point of view, it is plausible that new approaches, combined with silver-lean or silver-free materials, will enable a reduction in silver consumption toward a more ambitious target value of ∼1.0 mg/W_p, which may be necessary for long-term sustainable PV production at the terawatt level although significant technical and industrial challenges remain.

SUPPLEMENTARY MATERIAL

See the supplementary material for Table S1, which lists published screen- and stencil-printed finger widths for silicon solar cells from 2008 to 2025, together with the corresponding literature sources used to derive the learning curve shown in Fig. 10.

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AUTHOR DECLARATIONS

Conflict of Interest

The authors have no conflicts to disclose.

Ethics Approval

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Author Contributions

Andreas Lorenz: Conceptualization (equal); Data curation (equal); Formal analysis (equal); Investigation (equal); Methodology (equal); Visualization (equal); Writing – original draft (equal). **Baljeet Singh Goraya:** Conceptualization (equal); Data curation (equal); Formal analysis (equal); Methodology (equal); Supervision (equal); Visualization (equal); Writing – original draft (equal); Writing – review & editing (equal). **Sebastian Pingel:** Data curation (supporting); Formal analysis (supporting); Methodology (supporting); Validation (supporting); Writing – review & editing (supporting). **Sebastian Nold:** Conceptualization (supporting); Data curation (equal); Methodology (supporting); Supervision (supporting); Visualization (supporting); Writing – original draft (equal); Writing – review & editing (supporting). **Florian Clement:** Conceptualization (supporting); Funding acquisition (equal); Methodology (supporting); Project administration (equal); Supervision (supporting); Validation (supporting); Visualization (supporting); Writing – review & editing (supporting). **Ralf Preu:** Conceptualization (equal); Formal analysis (equal); Funding acquisition (equal); Methodology (equal); Project administration (equal); Supervision (equal); Validation (supporting); Visualization (equal); Writing – review & editing (equal).

DATA AVAILABILITY

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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